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Biopower

Securing American Leadership in Biotechnology

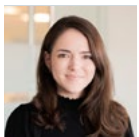
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The CNAS Technology and National Security Program produces cutting-edge research and recommendations to help U.S. and allied policymakers responsibly win and manage the great power competition with China over critical and emerging technologies. The escalating U.S.-China competition in artificial intelligence (AI), biotechnologies, next-generation information and communications technologies, digital infrastructure, and quantum information sciences will have far-reaching implications for U.S. foreign policy and national and economic security. The Technology and National Security Program focuses on high-impact technology areas with in-depth, evidence-based analysis to assess U.S. leadership vis-à-vis China, anticipate technology-related risks to security and democratic values, and outline bold but actionable steps for policymakers to lead the way in responsible technology development, adoption, and governance. A key focus of the Tech Program is to bring together the technology and policy communities to better understand these challenges and together develop solutions.

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Executive Summary

Cutting-edge biotechnologies will create organs, capture carbon emissions, restore polluted environments, tailor medicines to a person's genes, and replace vulnerable supply chains for food, fuel, fabrics, and firepower with domestic biobased alternatives.

The biorevolution is upon us. Converging breakthroughs in biological sequencing, engineering biology, and machine learning are ushering in an almost science fiction-like world in which humans can manipulate and even design the building blocks of life with increasing sophistication—for good or ill. In this world, cutting-edge biotechnologies will create organs, capture carbon emissions, restore polluted environments, tailor medicines to a person's genes, and replace vulnerable supply chains for food, fuel, fabrics, and firepower with domestic biobased alternatives. According to one estimate, existing biotechnologies could have a direct economic impact of \$4 trillion a year for the next 20 years.¹ As innovation continues, the ceiling could be far higher.

If next-generation biotechnologies hold great promise, they also come with gathering perils from new bioweapons, intrusive biosurveillance, and the race for biotechnology breakthroughs without adequate safeguards for public health, the environment, and democratic values. For policymakers, the question is not whether the biorevolution has transformative power, but which nation will responsibly harness that power to unlock new tools for defense, health, manufacturing, food security, environmental remediation, and the fight against climate change. No country is better positioned to lead the biorevolution than the United States, but it requires that policymakers act now with swift, ambitious, and far-sighted steps to secure America's place as the global biopower.

The United States enters the biorevolution with formidable tailwinds—an unrivaled innovation ecosystem, world-leading research institutes, unmatched private investment, talent, and a global network of democratic partners and allies. Recent federal investments and an

emerging policy framework have fortified U.S. leadership. But in this fast-moving field, settling for gradual progress will guarantee falling behind as competitors like China race to eclipse the United States with ambitions to scale up their biotechnology research, innovation, talent, and infrastructure.

To secure America's place as the global biopower, the next administration and Congress should accelerate U.S. tailwinds through greater investment in biotechnology research and infrastructure, especially in sectors beyond health and medicine; expand the pipeline of biotalent; and lead globally to drive biotechnology norms, standards-setting, and responsible adoption. At the same time, policymakers must navigate headwinds that could imperil further progress—specifically, an underdeveloped national biomanufacturing infrastructure; insufficient public and private investment that flows overwhelmingly to biotechnology research and development (R&D) in the health and medical sectors; a lack of uniform federal standards, definitions, and codes; a morass of conflicting policies and regulations; inaccessible and insecure biodata; and low public awareness and trust in emerging and ethically fraught biotechnology applications. This report outlines several recommendations to shore up America's position as the preeminent biopower, including an investment of \$20 billion in new federal funding. Policymakers should view this level of investment as the floor of what it will require to secure U.S. biotechnology leadership.

Regardless of what U.S. policymakers do, countries around the world are moving swiftly to embrace the biorevolution. The United Kingdom (UK) is driving innovation by concentrating and sharing its biodata through the UK Biobank, which houses the fully sequenced genetic codes of 500,000 people.² France has invested roughly \$9.5 billion through Innovation Santé 2030 to drive biomedical research.³ Japan has committed \$3 billion to promote its biotechnology ecosystem.⁴ South Korea is carving out a niche in digital biotechnology and aims to transition 30 percent of its manufacturing industry to biomanufacturing within a decade.⁵

If any nation can surpass the United States as the global biopower, it will be China. In its most recent five-year plans, Beijing made explicit its ambition to become a biotechnology “superpower.” It is well on its way. China's biotechnology leadership has surged on the back of significant public investment, long-term policy prioritization, a massive domestic market, decades of largely unrestricted capital flows, and the amassing of biodata through licit and illicit means.⁶ China's concerted biotechnology push has already paid dividends: its

scholars rank second in the world for authoring biomedical papers, and the country leads high-impact research in biofuels and biomanufacturing. China's high-impact research in synthetic biology is more than triple that of the United States, posing a high risk of monopolization.⁷ Today, China is a global biomanufacturing powerhouse that exports roughly 40 percent of the world's active pharmaceutical ingredients.⁸ Now, China aspires to move up the biotechnology value chain with a renewed push to support start-ups, integrate its vast biodata with cutting-edge machine learning tools, and dominate emerging markets for biotechnology with “national champions” such as BGI Group and WuXi Biologics, as it did with Huawei and 5G.

China's ambition to close the gap with the United States should inspire action from policymakers to secure and extend America's lead. **To that end, this report outlines a series of immediate and longer-term recommendations in six key areas for leaders in policymaking and industry:**



Invest in Research and Infrastructure

While the private sector largely drives biotechnology innovation, federal support is vital to general biotechnology R&D—especially in applications beyond health and medicine. To level up American biopower, policymakers should significantly expand public investment in biotechnology-related research, development, and infrastructure through the Department of Defense, Department of Energy, National Science Foundation, National Institutes of Health, and National Institute of Standards and Technology and support agency-led efforts to realize the March 2023 *Bold Goals for Biotechnology and Biomanufacturing*. Wherever possible, investments should seek to close the gap between the biomedical and biopharmaceutical sectors—which receive the lion's share of public and private investment—and other promising biotechnology applications in areas such as defense, energy, and manufacturing. At the same time, the federal government should better coordinate and strategically target existing and future investments in general and applied biotechnology research. Finally,

policymakers should consider financial and tax incentives to attract private capital to sustain biotechnology ventures across the Valley of Death, with a focus on biotechnology ventures in defense, manufacturing, energy, and agriculture.



Expand Biotalent

American biopower hinges on world-class biotalent in government, academia, and industry. Although the United States has a relative advantage in access to top STEM talent, it still faces a shortage that impedes progress. Industry should intensify engagement with secondary and postsecondary educational institutions to build early interest in biotechnology careers, expand internship and apprenticeship opportunities, and align curricula with anticipated biotechnology jobs. Policymakers should also pursue targeted immigration reform to supercharge the flow of top biotalent to U.S. research institutions and companies—paired with efforts to shore up research security—while acting to broaden biotalent and bioliteracy across government by leveraging initiatives such as the Presidential Innovation Fellowship and the U.S. Digital Service.



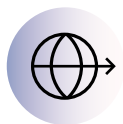
Unlock Biodata

Biotech leadership hinges on secure, quality biodata. However, biodata is too often segmented and sequestered in public and private labs according to different protocols with incompatible systems. Policymakers should clarify and modernize regulations for the use of artificial intelligence–ready health and genomic data for biotechnology research; invest in privacy-preserving and privacy-enhancing technology to promote large-scale biodata analysis while respecting privacy; establish a new American Biobank modeled on the successful UK Biobank; and fortify the cybersecurity of biodata repositories in government and private hospitals and clinics.



Strengthen and Streamline Policy

To the extent they exist, biotechnology standards, definitions, and regulations are outdated, inconsistent, and incoherent—sapping innovation. Policymakers should require a government-wide survey to identify regulatory gaps and develop uniform federal definitions, standards, and codes for biotechnology and the bioeconomy to improve tracking, measurement, and coordination. They should also establish a consolidated federal office to provide a single point of entry for biotechnology regulations and direct most federal agencies to develop strategies for aligning their various programs, investments, and outreach to drive responsible biotechnology innovation and broader biotechnology competitiveness.



Lead Globally

The United States should fully exercise its global leadership to shape the future of biotechnology. As with artificial intelligence (AI), Washington's decisions in the next few years could determine whether next-generation biotechnologies will reflect democratic values or techno-authoritarianism. To that end, policymakers should convene a global Biotechnology Safety and Opportunity Summit modeled after the success of the UK's 2023 Bletchley AI Safety Summit; set a global example by declaring red lines for unacceptable biotechnology research and applications; and develop a strategy for biotechnology standards-setting and joint research through existing partnerships such as the North Atlantic Treaty Organization (NATO); the Australia, United Kingdom, and United States trilateral security partnership (AUKUS); and the Quadrilateral Security Dialogue (the Quad).



Build Public Trust

Strong and sustained U.S. biotechnology leadership requires enduring support from policymakers and the public. However, both groups remain broadly unaware of the looming biorevolution and its consequences for America's economy, security, and values. To elevate and broaden the debate, industry should hold a national summit on the bioeconomy to make the case for why biotechnology leadership is essential to U.S. national, health, and economic security, and highlight biotechnology's positive applications. Senior executive branch leaders should crisscross the country and make the case for U.S. biotechnology leadership. For its part, Congress should hold regular biotechnology hearings and engage constituents to boost awareness among public and elected officials.

The biorevolution holds great promise and risk, and the ultimate balance will hinge largely on America's choices in the coming years. Policymakers must embrace the biorevolution with ambition and humility, acting boldly to drive research, innovation, and infrastructure across the full spectrum of biotechnologies; reform policy; train, attract, and retain biotalent; responsibly unlock biodata; leverage global partnerships; and earn public trust. The United States led the 20th century in biotechnology, and there is no reason it cannot lead the 21st as the preeminent global biopower.

Chapter 01

The Biorevolution

If the 20th century saw the Information Revolution, the 21st is witnessing the biorevolution—a dizzying future in which humanity can design and manipulate the building blocks of life with greater precision, for good or ill. Propelled by converging breakthroughs in synthetic biology, biological engineering, biocomputing, and more, the biorevolution will have far-reaching consequences for the global economic and security landscape, even if those impacts remain poorly understood by policymakers and the public.

The biorevolution rests on a powerful and unsettling truth: the building blocks of life, such as DNA and RNA, are programmable. With emerging biotechnologies and techniques, humans can now shape life for our ends with increasing sophistication—with the potential for better crops, livestock, materials, medicines, and manufacturing, along with powerful new tools to promote food and energy security, power the economy, safeguard the environment, and protect Americans from future pandemics and biothreats. Biology is transitioning from a longtime scientific discipline of observation and measurement to a domain of human design.⁹

The United States has pushed the frontier of biotechnology for decades. In the 1970s, American scientists developed recombinant DNA technology, unlocking medical advances such as monoclonal antibodies and insulin treatments for diabetes.¹⁰ In the 1980s, American biochemist Kary Mullis developed the polymerase chain reaction (PCR)—a foundational tool in modern genetics that allows researchers to reproduce DNA sequences.¹¹ Between 1990 and 2003, America spearheaded the Human Genome Project, a landmark scientific collaboration that generated the first nearly complete sequence of all three billion base pairs of the human genome.¹² A decade later, Jennifer Doudna earned a Nobel Prize

for helping to pioneer gene editing technology with CRISPR-Cas9.¹³ More recently, Drew Weissman helped lay the groundwork for the mRNA (messenger RNA) COVID-19 vaccine, which saved millions of lives.¹⁴

With emerging biotechnologies and techniques, humans can now shape life for our ends with increasing sophistication.

Although the United States led the world in biotechnology in the 20th century, there is no guarantee it will remain the global biopower in the 21st. America's competitors, especially China, grasp biotechnology's potential to enhance economic and military power and have moved proactively to realize it. In fact, China has already surpassed the United States in trials for CAR T-cell therapy—a promising cancer treatment.¹⁵ As the world's current biotechnology leader, the United States still enjoys the strongest hand to become the 21st century biopower, propelled by natural tailwinds from its innovative private sector, premier research institutions, and dynamic capital markets. These advantages are no reason for the United States to rest on its laurels. Ambitious competitors and gathering domestic headwinds risk seriously eroding U.S. biotechnology leadership without significant changes to policies and investments today.

At a time when many emerging technologies compete for limited attention and resources, there is value in clarifying the benefits of U.S. biotechnology leadership and the dangers of ceding it to competitors. This report will make the national and economic security case for U.S. biotechnology leadership; review the history, recent developments, and priority issues in U.S. biotechnology

policy; identify tailwinds and headwinds to U.S. biotechnology leadership; and assess biotechnology-leading nations abroad, from close partners to strategic competitors such as China. Finally, the report will outline recommendations in six key areas to secure America's place as the 21st century biopower.

Biotechnology, Biomanufacturing, and the Bioeconomy

Biotechnology's cross-cutting breadth, rapid advancements, and technical complexity can make it hard for policymakers and the public to grasp. In its simplest form, biotechnology is the manipulation of biology to create services and products. These range from precision medicine tailored to an individual's DNA to genetically engineered crops that better resist drought and disease to alternative proteins; mRNA vaccines; and biobased armor, resins, fabrics, and fuel. If "biotechnology" captures the range of technology-based techniques and tools, then "biomanufacturing" is its application to create specific products.¹⁶ By extension, the term "bioeconomy" captures the economic activity generated from every product, service, and process derived from

biological resources, even if they do not use biotechnologies.¹⁷ According to the National Academies of Science, Engineering, and Medicine (NASEM), in 2016, the bioeconomy accounted for more than 5 percent of the U.S. gross domestic product (GDP)—nearly \$1 trillion.¹⁸ An estimate from McKinsey & Company predicts that biological processes could generate up to 60 percent of all physical inputs in the global economy, with a direct economic impact of up to \$4 trillion annually between 2030 and 2040.¹⁹

In its simplest form, biotechnology is the manipulation of biology to create services and products.

Historically, discussion of biotechnology and the bioeconomy tends to focus on research, development, and applications in the health, medical, and pharmaceutical sectors—reflecting biotechnology's overwhelming share of public and private investment and activity. This report, however, will embrace a broader conception of biotechnology and the bioeconomy to reference the full spectrum of applications in health, energy, agriculture, and defense.



Biotechnology-enabled mRNA vaccines saved millions of lives in the United States and trillions of dollars in further economic harm. (Morsa Images via Getty Images)

The pandemic was a painful but powerful demonstration of biotechnology's power to save millions of lives and trillions of dollars.

The world has already reaped many benefits from the biorevolution, from genetically engineered crops that are hardier and more nutritious to stem cell treatments for leukemia and sickle cell anemia to advanced biofuels that have boosted energy security and reduced carbon emissions. It took the Human Genome Project 13 years and \$2.7 billion to sequence the human genome.²⁰ Today, cutting-edge machines can do this in less than a day, for less than \$1,000—and by the end of the decade, perhaps, for less than \$100.²¹ The COVID-19 pandemic exacted a terrible toll—more than 1.5 million American lives lost and \$5 trillion in federal spending—but it is worth imagining how much worse it would have been without biotechnology-enabled vaccines and the biotechnology-powered ability to sequence the virus within weeks.²² The pandemic was a painful but powerful demonstration of biotechnology's power to save millions of lives and trillions of dollars.

Key Definitions

Biotechnology: Technology that manipulates biology to create products and services, such as gene editing.

Biomanufacturing: The application of biotechnology-based techniques and tools to create specific products, such as biobased armor, fabrics, and fuel.

Bioeconomy: The economic activity generated from every product, service, and process derived from biological resources, even if it does not use biotechnologies.

Key Trends and Technologies

Although the biorevolution has gathered momentum for years, several converging trends will continue its acceleration: the collapsing cost of computing power, gene sequencing, and gene synthesis; the increasing sophistication and digitization of medical testing and records, which will dramatically expand biodata; and the rise of powerful artificial intelligence (AI) and machine learning tools. These trends have combined to give scientists and industries more biodata to analyze at lower cost and with greater power and precision.

These converging trends have powered biotechnology breakthroughs in several areas. Although detailing each breakthrough is beyond the scope of this report, policymakers should pay particular attention to developments in three key, often overlapping, areas: biological sequencing, engineering biology/synthetic biology, and human-machine interfaces.²³

Biological sequencing. This field includes efforts to map, measure, and analyze biological molecules and pathways. The most consequential subset is the sequencing of genes, the instructions for building and maintaining the cells in living organisms. When arranged in a sequence, genes form strands of DNA, one of the building blocks of life. Advances in biological sequencing therefore bestow greater insights into the design and function of life, with far-reaching applications, from understanding novel biothreats, such as the COVID-19 virus, to developing personalized medicine to designing new biobased materials and manufacturing processes. Other promising subfields include the sequencing of proteins, metabolites, the microbiome, and RNA transcripts—the molecules that help regulate the flow of genetic information within cells.

Engineering biology/synthetic biology. These terms are often used interchangeably, but they have important distinctions. Whereas engineering biology is the application of engineering principles to *existing* biological systems to develop products, synthetic biology uses modern biotechnologies and techniques to create *entirely novel* biological systems. For example, manipulating the DNA of a cotton plant would fall under engineering biology, while synthetic biology would entail designing the DNA of a new and improved cotton plant *de novo*.

Biological engineering spans revolutionary gene editing technologies from CRISPR and TALEN to the Pfizer-BioNTech and Moderna COVID-19 vaccines, which use mRNA technology.²⁴ Applications include lab-grown meat and bioengineered materials such as plant-based fibers, polymers, and coatings.

Engineering biology techniques are also increasingly used to enhance regenerative medicine, which involves repairing, replacing, or regenerating damaged cells, tissues, and organs.²⁵ Among the most promising advances in regenerative medicine is stem cell research, in which scientists use stem cells to 3-D print organic tissue and biobased materials.²⁶ In a decade or more, some experts envision a world of 3-D printed organs for human transplant.²⁷ In fact, British doctors have already successfully implanted a 3-D printed eye in a patient and 3-D printed bladders that saved the lives of three children.²⁸

Biological engineering and synthetic biology hold immense potential. Future breakthroughs could allow societies to complement chemical- and petroleum-based materials with bioengineered materials with a lower environmental impact, just as they could complement conventional, carbon-intensive meats with lab-grown alternatives. They could provide healthy lungs, livers, kidneys, eyes, and other organs to any patient who needs them, and replace or regenerate diseased cells to treat sickle cell anemia and cardiovascular disease.²⁹ None of these breakthroughs are imminent; indeed, some may be decades away. But given their potential, companies and countries would be unwise to ignore them.

Human-machine interfaces. This field sits at the intersection of organic biosystems (such as human organs) and inorganic systems (such as software and hardware). The most common applications are in the medical field, in which devices can create one- or two-way communications with parts of the body. For example, doctors can now place sensors on the scalp or implant them on the brain (“brain-computer interface”) to read a patient’s neural signals and control a robotic arm or a device cursor.³⁰ Other biomachines can send electronic signals to the brain (“deep brain stimulation”) to modulate unusual neural activity and help manage the symptoms of Parkinson’s disease, or to help treat attention deficit disorder, obsessive compulsive disorder, and severe depression.³¹ Similarly, a cochlear implant in the ear sends electronic signals to stimulate the auditory nerve and improve hearing.³² Although this field has seen less progress than biological sequencing and biological engineering, there is growing potential for medical applications. These include neuroprosthetics to restore lost senses through technologies such as bionic vision, or technologies to direct a prosthetic limb with signals from the brain.³³ Last year, Elon Musk’s company Neuralink made headlines for successfully implanting a chip onto a patient’s brain, reportedly restoring some mobility.³⁴ Neuralink was not the first successful implantation of

a neural chip, and it won’t be the last.³⁵ In September 2024, the U.S. Food and Drug Administration (FDA) cleared Neuralink’s Blindsight device, which Musk claims could one day restore limited vision to certain people who are blind, for accelerated development.³⁶ In time, biotechnology may be able to restore sight, hearing, and movement for millions—improving quality of life and productivity.

Developments in these three areas—biological sequencing, biological engineering/synthetic biology, and human-machine interfaces—will converge with other technology trends, especially the rapid expansion of advanced computing power and advanced AI tools and automation. According to AI research organization Epoch AI, from 1950 to 2010, it took almost two years to double computing power. But since 2010, it now takes only six months.³⁷ Today, computing costs are up to 1,000 times cheaper than they were two decades ago.³⁸ New machine learning techniques combined with the proliferation of biodata could allow researchers to extract transformative insights in health, medicine, agriculture, energy, and more.

Key Fields in Biotechnology

Biological sequencing: Efforts to map, measure, and analyze biological molecules and pathways. This includes sequencing not only genes but also proteins, metabolites, and RNA transcripts.

Engineering biology/synthetic biology: Engineering biology is the application of engineering principles to existing biological systems to develop products. Synthetic biology uses modern biotechnologies and techniques to create entirely novel biological systems.

Human-machine interfaces: This field sits at the intersection of organic biosystems, such as organs, and inorganic systems, such as software and hardware. This field includes devices that can create one- or two-way communication with parts of the body, such as Neuralink.

The National Security Case for U.S. Biotechnology Leadership

As with any technological breakthrough, bad actors will seize on the biorevolution's new tools and capabilities for their own nefarious ends. The same techniques that can sequence a deadly virus to develop a vaccine could also reveal how to make that virus even deadlier. The techniques to genetically manipulate staple crops to make them more resilient could yield insights about how to destroy them. The power to manipulate life inevitably unlocks the power to destroy it. Beyond direct threats to human security lie more fundamental questions about the dangers and ethics of “playing God” by treating DNA and RNA—the building blocks of life itself—as little more than computer code to manipulate for human ends.

Any qualms about the biorevolution should be tempered with the reality of its arrival. It is not a question of whether the biorevolution will happen, but which nations will lead it, shape it, and reap its benefits. The United States continues to lead the world in biotechnology, but China races to close the gap. Beijing has named biotechnology a strategic emerging industry, and it has already hit its 10-year targets. According to some estimates, China has invested up to \$100 billion in life sciences research and development (R&D) at the local, provincial, and national levels.³⁹ Leading Chinese biotechnology companies are also amassing vast genomic and health data, and the government's prioritization of the sector has offered a strong signal for research and market investment.⁴⁰

It is not realistic for the United States to out-compete China in every sector. Federal actions to protect and support certain sectors require a clear-eyed assessment of their importance to national security. The national security case for U.S. biotechnology leadership rests on five core arguments.

First, a strong biotechnology sector will better protect the nation's citizens and economy from new pandemics and biothreats—both natural and human created. The COVID-19 pandemic painfully underscored the danger of unchecked viruses to the safety of America's citizens and economy. A robust biotechnology sector will put the United States in a stronger position to anticipate, prevent, and mitigate future outbreaks and biothreats with the latest tools for virus detection, sequencing, and vaccine development. To weigh the value of a robust domestic biotechnology sector, consider a world in which the United States must rely on China for life-saving vaccines for the next pandemic or a neglected U.S.

biotechnology sector can only produce those vaccines months—or even years—later than foreign competitors. Beyond this, biotechnology has the potential to deliver breakthroughs to restore vision, sight, and movement through neuroprosthetics; tailored wellness and dieting regimes through personalized medicine; and precision treatments for sickle cell anemia and cardiovascular disease to improve the longevity, health, and productivity of Americans. Embracing the biorevolution could also avoid a future in which the United States must rely on competitors such as China for mission-critical materials for the U.S. military, clean energy breakthroughs, or cutting-edge tools to protect its population from novel pandemics and biothreats.⁴¹

The power to manipulate life inevitably unlocks the power to destroy it.

Second, ceding the biorevolution to competitors risks depriving the United States of potentially transformative defense applications. Breakthroughs in biotechnology could provide renewable sources of jet fuel.⁴² They could promote the physical and mental health of service members through products that help balance their microbiomes, enable self-repairing airfields with “living concrete” full of bacteria that regenerate to fill cracks as they emerge, and better protect soldiers with armor made of “dragon silk”—a genetically engineered spider silk that is not only lighter but up to three times tougher than traditional Kevlar.⁴³ The U.S. Department of Defense (DoD) could employ new, lighter, and flame-retardant biological resins to improve the designs of drones, ships, and aircraft.⁴⁴ Biological sensors could provide real-time alerts to hazardous biological or chemical agents.⁴⁵ Researchers have manipulated the *E. coli* bacteria into a biosensor that can detect unexploded ordnance, reducing risks to both service members and civilians.⁴⁶ Biotechnology's defense applications are as far ranging as the field itself.⁴⁷

Third, the biorevolution will bestow significant economic benefits to nations that harness it. Like AI, biotechnology could level up major sectors of the economy—from agriculture and pharmaceuticals to textiles, energy, and manufacturing. If the United States fails to invest in biotechnology, it risks missing out on the new tools, processes, and productivity it could unlock—and the billions, even trillions, of dollars in economic benefit. McKinsey & Company forecasts that *existing* biotechnology applications could have up to \$4 trillion in annual, direct economic impact over the next 20 years.⁴⁸



A recycling center converts food waste into biomass—a source of renewable energy. (Justin Paget via Getty Images)

In addition, the World BioEconomy Forum estimates that the broad adoption of synthetic biology techniques in manufacturing could account for more than one-third of all global output—or nearly \$30 trillion.⁴⁹

Biotechnology can also strengthen domestic supply chains, for example, through biologically engineered crops that increase food security, a robust bioproduction capacity to strengthen national security supply chains, and a strong biopharmaceutical sector to ensure access to breakthrough vaccines and treatments in the event of new pandemics and biothreats.⁵⁰ The biorevolution could also enable more resilient supply chains that promote broad-based economic growth. Today, manufacturing processes often rely on imported chemical inputs, exposing them to external supply chain shocks. By contrast, biomanufacturing relies on organic biomass ranging from agricultural byproducts such as animal waste to forestry byproducts like bark and wood chips. Unlike imported chemical inputs, these bio-inputs are renewable, domestically sourced, and broadly distributed

across the country. As such, biomanufacturing facilities could be colocated with farms, forests, and municipal waste sites across the country, driving new jobs and investments in more remote and rural communities.

Fourth, biotechnology could unlock powerful tools to fight climate change and promote energy security. Climate change threatens U.S. national security in myriad ways, from direct harm to citizens from extreme weather to increased geopolitical instability from persistent drought and extreme heat.⁵¹ As Secretary of Defense Lloyd Austin has said, “No nation can find lasting security without addressing the climate crisis.”⁵² Fortunately, experts predict biotechnologies could someday yield advanced biofuels to significantly reduce emissions in transport, industry, and agriculture.⁵³ Organic biomass—everything from fallen trees to food waste to farm products—is an abundant, renewable energy source. However, converting biomass to fuel still requires an expensive process of breaking down the cell walls into sugars and converting those sugars into

fuel. Biofuel proponents hope advances in synthetic biology can streamline this process and make biofuels a more compelling alternative. Biotechnology also holds promise for carbon reduction and sequestration—from carbon-absorbing microalgae to genetically modified trees that pull more carbon dioxide from the air to genetically modified crops that require less energy-intensive fertilizers.⁵⁴ As the urgency of tackling climate change and achieving energy security grows, so does the case for developing biotechnology-powered tools to advance these goals.

Biotechnology can also transform industrial processes to become less carbon intensive. For example, BDO (1,4-Butanediol) is a widely used chemical for manufacturing plastics and fibers. Companies such as Genomatica are developing biobased alternatives to BDO that use the fermentation of renewable feedstocks, such as sugarcane, instead of conventional chemical inputs. According to Boston Consulting Group, if the world's 30 BDO manufacturers adopted bio-BDO, it would prevent the production of more than 15 million tons of carbon emissions annually—equivalent to the emissions of a million Americans.⁵⁵

Fifth, the United States has a strategic opportunity and moral responsibility to set global norms and standards for biotechnology consistent with democratic values. The biorevolution raises difficult questions related to privacy, ethics, safety, and responsible biotechnology research, development, and deployment. The United States and China have fundamentally divergent views and practices surrounding the privacy of health data and the ethics of biological engineering more broadly. Infamously, Chinese scientist He Jiankui used CRISPR to alter three human embryos, two of whom were born in 2018 as Lulu and Nana, and a third who was born in 2019—the first genetically modified humans in history.⁵⁶ Although China has since established stronger ethics rules for human studies, they completely exempted the private sector—a troubling omission.⁵⁷

Just as troubling, the Chinese government has shown few qualms about using biotechnology for population control and repression of its ethnic and religious minorities, such as the Uyghurs. In the Xinjiang region, Chinese scientists are reportedly trying to use DNA samples to create an image of a person's face, with which it could presumably use to track and target them.⁵⁸ The government also appears to have embraced a troubling vision for offensive biotechnology capabilities.⁵⁹ In December 2021, the U.S. Department of Commerce (DOC) placed several Chinese research institutes and biotechnology firms on the entity list for helping the Chinese military

develop purported “brain control” weapons.⁶⁰ In the past, Chinese military leaders have openly speculated about biotechnology's potential offensive uses, for example, by enabling “specific ethnic genetic attacks.”⁶¹ China's troubling biotechnology record makes it especially urgent for the United States to proactively shape global biotechnology norms and standards.

Given the stakes, securing the United States' position as the 21st-century biopower warrants greater attention and action from policymakers to reinforce existing advantages, shore up vulnerabilities, and seize opportunities. To that end, chapter two of this report surveys the current state of U.S. biotechnology leadership, focusing on private sector strengths and weaknesses, along with a deeper dive into federal government policies, investments, and initiatives. It identifies financial, policy, regulatory, workforce, and other headwinds to U.S. biotechnology leadership. In chapter three, this report looks abroad to survey leading biotechnology allies, partners, and competitors to identify their relative strengths and weaknesses. **Drawing on these assessments, the authors outline immediate and longer-term recommendations for leaders in policymaking and industry in six key areas:**

1. Invest in research and infrastructure
2. Expand biotalent
3. Unlock biodata
4. Strengthen and streamline policy
5. Lead globally
6. Build public trust

Chapter 02

The State of American Biopower

This report will assess the state of American biopower by examining specific tailwinds and headwinds to U.S. leadership. The tailwinds emanate from broader structural strengths of the U.S. economy, as well as specific policy choices in the past half century. The headwinds, by contrast, are largely the result of inadequate policy and investment, as well as a rising challenge from foreign competitors. This section will briefly review key developments in U.S. biotechnology policy and survey the most important headwinds and tailwinds to American biopower.

U.S. Biotechnology Policy from Reagan to Biden

U.S. biotechnology policy remains a work in progress. To date, the United States has no single major law providing specific regulation for biotechnologies.⁶² Rather, federal regulation falls primarily within the cross-cutting jurisdictions of the FDA, the Environmental Protection Agency (EPA), and the United States Department of Agriculture (USDA), reflecting the technology's diffuse, general-purpose nature. Early on, federal regulation emphasized a risk-based approach to ensure protection for public safety, health, and the environment from biotechnology products, while maintaining sufficient flexibility to promote innovation and competitiveness. It proved a difficult balance to strike, especially as the three leading agencies independently developed their own rules, procedures, and regulations.

Over time, the overlapping jurisdictions for biotechnology created regulatory confusion. In response, the

Reagan administration created a Coordinated Framework for Regulation of Biotechnology in 1986 to clarify regulatory responsibilities across the federal government.⁶³ The first Bush administration shifted the framework to a more “process agnostic” approach that focused more on a biotechnology product’s application and context, rather than assuming it was inherently riskier because it was derived from biotechnologies.⁶⁴

For more than two decades, federal biotechnology regulations remained largely unchanged, even as the sector notched major milestones—from Dolly, the first-ever mammal cloned from an adult somatic cell in 1996, to the development of the CRISPR platform in 2012.⁶⁵ The most important development of this period was passage of the Health Insurance Portability and Accountability Act of 1996, or HIPAA.⁶⁶ Although the law did not directly regulate biotechnologies, it created strict rules on one of its most important inputs: biodata. HIPAA established limits on the collection, storage, use, transfer, and disclosure of private health information (PHI), with the goal of protecting patients’ medical privacy.⁶⁷ Although HIPAA allows access to PHI for research, it narrowly defines research and establishes a multipart test to determine data’s eligibility for use. Although most biotechnology companies are not directly covered by HIPAA, the law can indirectly shape their access to biodata by regulating the covered health providers allowed to collect and share it.⁶⁸ HIPAA’s “minimum necessary standard,” for instance, governs the extent to which researchers can use patients’ health and genomic data. The standard guards against excessive data access to ensure patient privacy, but there remains considerable ambiguity over how it applies to genomic data, often impeding research.⁶⁹

Efforts to modernize and streamline biotechnology regulations continued in the Obama administration. The administration recognized that the overlapping regulatory structure had created needless obstacles that inhibited biotechnology innovation; for example, by creating uncertainty over agency jurisdiction and review timeframes. The complex regulatory structure proved especially hard to navigate for small and mid-sized companies and academic researchers. As a result, the White House updated the Coordinated Framework in 2015 to clarify existing roles and responsibilities across the three implementing agencies, among other provisions.⁷⁰ Separately, the Obama administration also saw the Defense Advanced Research Projects Agency (DARPA) open the Biological Technologies Office (BTO)—one of the DoD’s first forays into harnessing biotechnology for national security.⁷¹ The BTO focuses principally on supporting breakthrough capabilities for operational biotechnology, such as biosensors and biobased materials, and boosting warfighter readiness through improved tactical care and medical countermeasures.

Stepping back, the Trump administration continued a shift in federal policy from a more risk-averse emphasis on consumer safety, public health, and environmental protection to a focus on innovation and competitiveness.

During the Trump administration, U.S. leadership in critical and emerging technologies assumed a more central role in the broader U.S.-China competition. As with other emerging technologies, such as AI, the administration embraced a light-touch regulatory approach for biotechnology that emphasized innovation and private sector leadership. The June 2019 Executive Order on Modernizing the Regulatory Framework for Agricultural Biotechnology Products epitomized this approach.⁷² It prioritized innovation and economic growth and again directed the three coordinating agencies to streamline biotechnology regulations.⁷³ It also called for a unified online portal of federal biotechnology regulations, which the USDA now hosts.⁷⁴

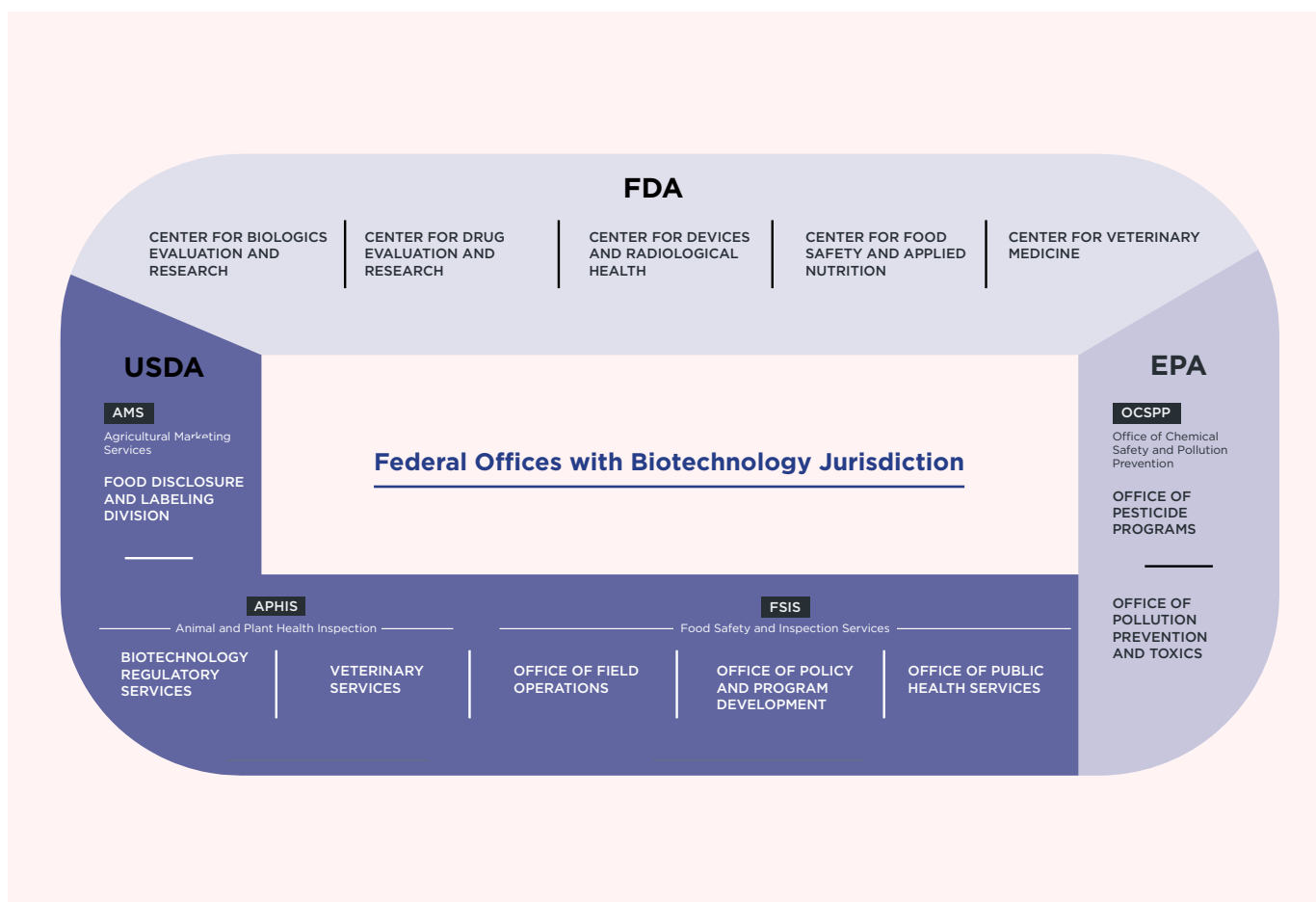
The Trump administration also laid the groundwork for one of the most significant biotechnology efforts to date: the Bioindustrial Manufacturing and Design Ecosystem (BioMADE), which launched in April 2021.⁷⁵ With an initial \$87 million investment from the DoD,

BioMADE was created as an independent, nonprofit body with a mission to foster a “sustainable, domestic end-to-end bioindustrial manufacturing ecosystem.”⁷⁶ It emphasizes collaboration between industry, academia, and government and supports initiatives to bridge the gap between “lab-scale research and at-scale manufacturing,” accelerate technology deployment or redevelopment, derisk key infrastructure, and promote education and workforce development.⁷⁷

Stepping back, the Trump administration continued a shift in federal policy from a more risk-averse emphasis on consumer safety, public health, and environmental protection to a focus on innovation and competitiveness. The Biden administration doubled down on this shift, but went much further to marshal federal tools, resources, and policy to catalyze U.S. biotechnology leadership. It also restored robust federal funding for R&D at the National Science Foundation (NSF), the National Institutes of Health (NIH), and the Department of Energy (DOE)—the historic foundation of biotechnology breakthroughs.

An early initiative was the June 2021 launch of the DOE’s Energy Earthshots Initiative, which seeks to catalyze clean energy breakthroughs within the decade.⁷⁸ Several Earthshots call for breakthroughs in biotechnologies. The Industrial Heatshot, for example, seeks to reduce heavy industry emissions by 85 percent by 2035 through supporting solutions such as “biobased manufacturing, electrolysis, ultraviolet curing, and advanced separations.”⁷⁹

The capstone of the Biden administration’s efforts was the September 2022 Executive Order on Advancing Biotechnology and Biomanufacturing Innovation for a Sustainable, Safe, and Secure Bioeconomy (Biotechnology Executive Order).⁸⁰ The Biotechnology Executive Order (EO) is the United States’ most ambitious and comprehensive vision for biotechnology leadership to date, with a clear emphasis on promoting and protecting the nascent sector and unlocking biotechnology’s benefits across the economy. The EO sought to galvanize agencies and offices across the federal government to place renewed emphasis on promoting and securing U.S. biotechnology leadership through a range of investments, actions, studies, and policy changes.⁸¹



The executive order's stated goals were principally to:

- bolster and coordinate federal investment in biotechnology R&D;
- cultivate a biological data ecosystem to promote innovation;
- improve and expand domestic biomanufacturing production capacity;
- boost domestic biomass production;
- expand market opportunities for bioenergies and biobased products;
- train and support a diverse and skilled biotechnology workforce;
- clarify and streamline regulations; and
- strengthen international cooperation around biotechnology R&D.

One of the EO's deliverables was a report on the state of biotechnology regulation under the Coordinated

Framework.⁸² Underscoring the overlapping and often burdensome regulatory structure for biotechnology, the report found that 13 different federal offices had jurisdiction over some element of the biotechnology sector, with six at the USDA, five at the FDA, and two at the EPA.⁸³

Arguably, the Biotechnology EO's most important step was mobilizing more than \$2 billion for various initiatives:⁸⁴

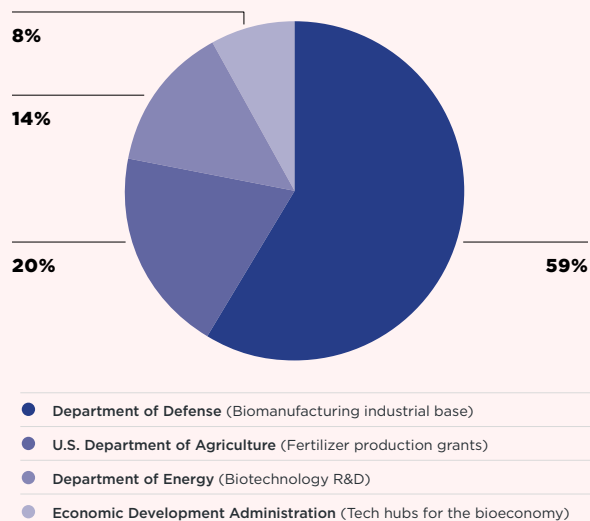
- **\$1.5 billion for the DoD** to jumpstart a domestic industrial base for biomanufacturing
- **\$500 million from the USDA** to support American innovative fertilizer production with five-year grants of up to \$100 million
- **\$358 million from the DOE** for biotechnology-related R&D for bioproducts and biomaterials
- **\$200 million from the U.S. Economic Development Administration** to create "tech hubs" to advance the bioeconomy

Although the Biotechnology EO was well received among biotechnology advocates in government and industry, implementation has been slow and uneven. The Federation of American Scientists developed a tracker of all the deliverables required by the 2022 Biotechnology EO, and, as of November 2024, more than half had a status of “unknown,” even though many were past their required deadline.⁸⁵

A major outcome of the 2022 Biotechnology EO was the March 2023 release of the *Bold Goals for U.S. Biotechnology and Biomanufacturing*.⁸⁶ Under the *Bold Goals*, five federal agencies identified priorities for an ambitious 20-year biotechnology research agenda.⁸⁷

1. **Climate (DOE):** Demonstrate and deploy biobased alternatives that could replace more than 90 percent of existing plastics and other commercial polymers.
2. **Food and Agriculture (USDA):** Reduce agricultural methane emissions to help reduce U.S. greenhouse gas emissions by 50 percent.
3. **Supply Chain (DOC):** Meet at least 30 percent of domestic demand for chemicals through competitive, sustainable biomanufacturing means.

2022 Biotechnology Executive Order Funding to Advance the Bioeconomy



Source: Bold Goals for U.S. Biotechnology and Biomanufacturing: Harnessing Research and Development to Further Societal Goals, The White House Office of Science and Technology Policy, March 2023, <https://www.whitehouse.gov/wp-content/uploads/2023/03/Bold-Goals-for-U.S.-Biotechnology-and-Biomanufacturing-Harnessing-Research-and-Development-to-Further-Societal-Goals-FINAL.pdf>.

4. **Health (U.S. Department of Health and Human Services):** Increase manufacturing of cell-based therapies, thereby expanding access, reducing health disparities, and decreasing the manufacturing cost 10-fold within 20 years.
5. **Cross-Cutting Advances (NSF):** Sequence the genomes of one million microbial species and clarify the function of at least 80 percent of these newly discovered genes within five years.

The *Bold Goals* offer a high-level vision to galvanize public, private, and academic investment and collaboration to realize biotechnology’s potential for addressing urgent national priorities. In that sense, it was also part of both the Trump and Biden administrations’ efforts to shift the public conversation—and federal bureaucracy—toward positive applications of biotechnology. However, the *Bold Goals* do not include any new funding or commitments to prioritize existing funds or efforts for biotechnology R&D. They simply point out the need to do so.

The Biotechnology EO also accelerated DoD efforts to boost the bioeconomy through two major new initiatives. First, it directed the DoD to create a Tri-Service Biotechnology for a Resilient Supply Chain (T-BRSC) program to provide more than \$270 million over five years to accelerate the R&D of defense-relevant supply chains, including “fuels, fire-resistant composites, polymers and resins, and protective materials.”⁸⁸ Second, it provided \$1 billion over five years for a new Distributed Bioindustrial Manufacturing Investment Program (DBMIP), which will expand domestic biomanufacturing capacity for products with both defense and commercial applications.⁸⁹ Unlike the conventional defense industrial base, there is no robust U.S. biomanufacturing industrial base today. The DoD’s efforts through T-BRSC, DBMIP, and its investments in BioMADE all seek to build this biomanufacturing industrial base from the ground up. Although there is significant overlap in the three initiatives’ broader goals, at a high-level, T-BRSC seeks to boost biotechnology R&D for *defense-specific* products and supply chains, whereas DBMIP can also support products with broader commercial applications, such as critical chemicals. BioMADE, by contrast, aims to foster America’s bioindustrial manufacturing base more broadly and focuses more on promoting cross-sectoral collaboration. In March 2023, the DoD increased BioMADE’s budget to \$450 million.⁹⁰

Since the Biotechnology EO, civilian agencies have launched many initiatives to boost American biopower. NSF created the BioFoundries Program to help build the



A Texas petrochemical plant releases emissions. The Department of Energy's Industrial Heatshot seeks to reduce heavy emissions by 85 percent by 2035 through unlocking the potential of biotechnologies. (Bloomberg Creative via Getty Images)

national infrastructure to drive U.S. biotechnology and the bioeconomy.⁹¹ NSF envisions these biofoundries as integrated facilities that combine biological disciplines with automation, AI, and advanced computing to drive biotechnology innovation, and it will award up to \$24 million per project to support biofoundries nationwide.⁹² In January 2024, NSF also designated one of its first “regional innovation engines” in North Carolina to drive breakthroughs in regenerative medicine and could invest up to \$160 million in the project over 10 years.⁹³

In October 2023, the administration also designated 11 tech hubs for biotechnology (out of a total of 31), supporting activity across 13 states and Puerto Rico.⁹⁴ Congress created the tech hubs initiative through the 2022 CHIPS and Science Act to encourage collaboration among industry, academia, and state and local governments to spur job creation, investment, and competitiveness in key technologies.⁹⁵

Notably, Congress also authorized the creation of a National Security Commission on Emerging Biotechnology (NSCEB) in the FY22 National Defense Authorization Act.⁹⁶ Inspired by the success of the National Security Commission on AI, Congress charged the NSCEB with reviewing how advancements in biotechnology will shape existing and future defense activities and U.S. national security more broadly. In January 2024, the commission released its *Interim Report*, which emphasized the potential of biotechnology to deliver breakthrough solutions in several previously discussed areas, from climate change to defense, while also underscoring the urgency of securing U.S. biotechnology leadership as competitors seek to close the gap.⁹⁷ The NSCEB plans to submit its final report in early 2025 and will operate through summer 2026 to engage policymakers and the public about its recommendations.⁹⁸

American Biopower: Tailwinds

No country boasts a stronger biotechnology sector than the United States, which has blazed the trail for the past half century. Like many other critical and emerging technologies, biotechnology benefits from America's powerful tailwinds. These include:

1. **World-class research institutions**
2. **Top global talent**
3. **Agile capital markets**
4. **Attractive intellectual property protections**
5. **Risk-tolerant innovation culture**
6. **Robust public and private funding for general and applied research**

Academic & Research Institutions

World-leading academic institutions are a pillar of American biopower. According to the U.S. News & World Report, 6 of the top 10 best global universities for biotechnology and applied microbiology are in the United States, including Harvard University, the Massachusetts Institute of Technology, Stanford University, the University of California at San Diego, Johns Hopkins University, and the University of Pennsylvania. Three are in China.⁹⁹ Biotechnology also benefits from significant upstream investments in biosciences by U.S. academic institutions. In 2020, U.S. academic R&D for bioscience was more than \$51 billion, up 28 percent since 2015.¹⁰⁰ Since 2019, however, annual growth for academic R&D bioscience spending has been just 2.8 percent—essentially flat in real terms.¹⁰¹

World-leading academic institutions are a pillar of American biopower.

According to the Australian Strategic Policy Institute (ASPI), the United States leads in three of seven key biotechnologies: vaccines and medical countermeasures, genetic engineering, and nuclear medicine and radiotherapy.¹⁰² China leads in the remaining four: synthetic biology, biological manufacturing, genomes and genetic sequencing and analysis, and novel antibiotics and antivirals.¹⁰³ The United States dominates high-impact research for vaccines and medical countermeasures.¹⁰⁴

This leadership was on full display during the pandemic, when the U.S.-based Pfizer-BioNTech and Moderna vaccines proved roughly 95 percent effective in reducing severe cases, compared to just 51 percent effective for China's CoronaVac vaccine, according to clinical data.¹⁰⁵

Patents, Publications, and Prizes

The United States continues to lead in high-quality research and patents related to biotechnology. Since 2012, the United States has generated far more highly cited biological publications—roughly 600 per year—than competitor nations, although China is quickly closing the gap, doubling its highly cited publications from roughly 200 in 2018 to 400 in 2021.¹⁰⁶ Between 1999 and 2013, U.S. patents also comprised up to 45 percent of all leading patents filed in biotechnology, medical technology, and pharmaceuticals.¹⁰⁷

The United States also leads in prestigious prizes in key fields for biotechnology. Since 1985, nearly half of Japan's International Prize for Biology awards have gone to American scientists. The United States likewise boasts the most Nobel Prize winners for chemistry, accounting for 76 out of 194 Nobel laureates from 1901 to 2023 for advancements in everything from developing new methods of gene editing to conducting the directed evolution of enzymes.¹⁰⁸

Public Investment

Robust and relatively stable federal funding has been essential to U.S. biotechnology R&D. Federal funding has principally supported basic R&D in biological sciences, with applied biotechnology R&D falling largely to the private sector.¹⁰⁹ Several federal agencies have funded biotechnology R&D and other initiatives and investments have predominantly flowed from NIH and NSF.¹¹⁰ As such, public investment has overwhelmingly supported biotechnology R&D related to medicine and health, dwarfing support for other promising fields such as energy and agriculture.

NIH is the world's largest public funder of biomedical research, investing most of its annual appropriations—nearly \$48 billion for FY23—in health and life sciences research, including for breakthrough medical treatments.¹¹¹ NIH funding has driven biotechnology progress via two pathways: first, through direct investment in biotechnology techniques such as DNA sequencing and protein engineering; and second, through broader investment in foundational research areas such as biophysics, virology, pharmacology, and molecular and cellular biology.¹¹²



A scientist loads a DNA gel to better analyze DNA fragments, demonstrating an important technique in molecular biology and genetic research. The National Science Foundation plays a vital role in funding general and applied biotechnology research, which is foundational to broader biotechnology innovation and the bioeconomy. (Assembly via Getty Images)

Over the years, these investments have had considerable spillover benefits for biotechnology. Arguably, the most seminal breakthrough came in the early 1960s, when NIH-funded researchers cracked the genetic code—transforming the foundational understanding of biology and opening the door to new research and innovation across a range of scientific disciplines.¹¹³ Following the 1961 discovery of mRNA, more than 50 years of NIH-funded research into viruses and vaccine delivery cleared the way for the development of the Pfizer-BioNTech and Moderna mRNA COVID-19 vaccines within the first 100 days of the pandemic, saving millions of lives and trillions of dollars.¹¹⁴

NSF has also driven U.S. biotechnology leadership through decades-long investments in fundamental research at the intersection of biology and engineering, computing, and other physical sciences; application-driven research to develop biobased materials, machines, and manufacturing technologies;

The USDA has also been central to driving U.S. biotechnology leadership, with a focus on research and applications related to forestry, farming, and growing the domestic market for biobased products.

education and workforce development; and partnerships with the private and nonprofit sectors to promote commercialization.¹¹⁵ NSF's interdisciplinary investments are well suited to biotechnology, which sits at the crossroads of biology and several other fields from engineering to mathematics to computer science. NSF invests roughly \$250 million annually to support U.S. biotechnology and the bioeconomy, and it was the first federal agency to invest in a synthetic biology research center.¹¹⁶ Today, NSF has more than 10 directorates and offices supporting biotechnology R&D.¹¹⁷

The USDA has also been central to driving U.S. biotechnology leadership, with a focus on research and applications related to forestry, farming, and growing the domestic market for biobased products. Often with support from the USDA, U.S. farmers have eagerly adopted genetically engineered crop varieties: in 2012, around 90 percent of all corn, cotton, and soybeans planted in the United States were produced through genetic engineering.¹¹⁸ The United States is the world's largest exporter of agricultural products, and biotechnology has expanded its trade balances in agriculture and helped feed the world. The USDA has also grown the domestic market for biobased products through its BioPreferred program, which boosts the purchases of biobased products—those derived from renewable organic materials such as plants and other agricultural, marine, and forestry materials—by requiring that federal agencies and contractors give purchasing preference to biobased products.¹¹⁹

Although NIH, NSF, and the USDA have been the primary drivers of federal biotechnology investment, other agencies have played a role through smaller investments or policy developments, including the FDA, EPA, DoD, DOE, and DOC.

Over the past quarter century, the White House and Congress have also worked together to drive signature initiatives that helped catalyze biotechnology breakthroughs. Most prominent was the Human Genome Project, which began in 1990 with a \$2.7 billion investment from Congress.¹²⁰ A later study found that for every \$1 invested in the project, it returned \$141 to the U.S. economy.¹²¹

Another initiative was the President's Emergency Plan for AIDS Relief (PEPFAR), which has invested more than \$100 billion since 2003 to strengthen the global response to HIV/AIDS.¹²² The historic investment not only saved more than 25 million lives but also helped lay a biotechnology foundation of laboratories, testing sites, expertise, and research breakthroughs that helped enable the rapid U.S. response to COVID-19 under Operation Warp Speed (OWS).¹²³ Launched under the Trump administration, OWS invested \$18 billion to accelerate private sector development and deployment of COVID-19 vaccines in less than a year.¹²⁴ Beyond preventing 18.5 million hospitalizations and up to 3.2 million additional deaths, OWS also significantly increased research in mRNA technologies and demonstrated the power of ambitious public-private partnerships to streamline regulation and jumpstart innovation.¹²⁵

The Human Genome Project, PEPFAR, and OWS all underscore the potential of well-resourced, ambitious public investments to deliver life-changing biotechnology

breakthroughs. However, all three initiatives focused on the health and medical sectors, again underscoring the historic underinvestment in the broader bioeconomy.

Private Investment

Significant private investment—overwhelmingly in the biomedical and biopharmaceutical sectors—remains an enduring asset for American biopower. During the COVID-19 pandemic, private investment peaked as the lifesaving vaccines demonstrated the sector's potential. In 2021, there were a record 114 biotechnology initial public offerings (IPOs) totaling \$16.1 billion in value, compared to just 47 IPOs totaling \$5.6 billion in 2019.¹²⁶ However, private biotechnology investment has since dropped precipitously. In 2023, there were just 11 biotechnology IPOs, totaling \$2.52 billion, on track to fall short of even prepandemic levels.¹²⁷ As of the third quarter of 2024, the S&P Biotechnology Select Industry Index was down more than 40 percent compared to its peak in February 2021.¹²⁸

U.S. biotechnology firms continue to win the race for global investment.

As public investments through short-term initiatives such as OWS ramped down, venture capital (VC) funding has helped sustain biotech innovation. In 2023, VCs invested nearly \$30 billion for biotechnology firms, down from a height of \$54.4 billion in 2021, but still more than double the prepandemic total of \$12 billion in 2019.¹²⁹

Focusing on the total volume or value of IPOs or VC funding, however, can obscure where the investment has flowed, and how that may shape the future of the U.S. biotechnology sector. In general, biotechnology firms have relatively high R&D intensity, longer product development cycles, and lower prospects of commercial success compared to, for instance, many narrow AI applications.¹³⁰ Compounding the challenge, uncertainty over additional federal funding and higher interest rates has made private capital more risk averse, pushing it toward more established areas and biotechnology firms and applications, primarily in medicine and biopharmaceuticals.¹³¹ This does not bode well for earlier-stage biotechnology start-ups in other fields, such as defense, energy, agriculture, and manufacturing.

In 2022, more than \$15 billion in VC funding flowed to platform biotechnologies—general biotechnology tools, such as CRISPR—with the rest flowing mostly to asset-based biotechnologies—narrower products or treatments

for specific purposes, such as insulin treatment for diabetes.¹³² VC funding for platform biotechnologies went largely to regenerative medicines, integrating large omics datasets, and new gene therapies, such as improved mRNA technology.¹³³ However, there is growing private sector interest—especially from large pharmaceutical companies and VC firms—in integrating AI and machine learning techniques into biotechnology. The biggest recipient of VC investments in platform biotechnologies between 2019 and 2022 was machine learning-enabled drug discovery, receiving more than \$9 billion.¹³⁴

Stepping back, U.S. biotechnology firms continue to win the race for global investment. In the first three quarters of 2023, 9 of the top 10 global funding rounds went to U.S. biotechnology companies. Only one of the companies was based in China (GenScript ProBio).¹³⁵ Again, private investment and activity in biotechnology continues to focus predominantly on the health, medical, and pharmaceutical sectors, at the expense of promising applications in areas like energy, manufacturing, and defense.

Another American strength is the relatively supportive environment for biotechnology start-ups. As of December 2022, the United States was home to 9 of the top 10 biotechnology accelerators worldwide.¹³⁶ As of 2021, the United States also had 2,840 active biotechnology companies—more than any other Organisation for Economic Co-operation and Development (OECD) country—followed closely by the United Kingdom (UK) with 2,770 biotechnology firms (as of 2023).¹³⁷ OECD data shows that the U.S. biotechnology sector also scores well on “R&D intensity,” a measure of the sector’s total R&D expenses as a share of its value added to the economy.¹³⁸ This allows for a rough comparison of how innovation oriented biotechnology sectors are across countries. By this measure, the United States lagged only Belgium, Switzerland, and Lithuania.¹³⁹

Bioscience Industry

The United States boasts a large and dynamic bioscience sector.¹⁴⁰ In 2021, the industry’s total direct economic impact was \$1.2 trillion, for a total \$2.9 trillion in indirect impact.¹⁴¹ According to a 2022 report from the Biotechnology Innovation Organization (BIO), the domestic bioscience industry employed 2.1 million people across more than 127,000 businesses, with employment growth outpacing the broader economy.¹⁴² Between 2002 and 2022, the number of life sciences researchers in the United States increased by 87 percent, compared to a 14 percent increase in jobs across the economy over the same period.¹⁴³ As of 2021, U.S. bioscience workers earned roughly \$126,000 a

year, 85 percent more than their average private sector counterparts.¹⁴⁴ Notably, the bioscience industry is also broadly distributed across the nation, with over half of all U.S. metropolitan areas boasting a concentration of employment in at least one bioscience industry subsector or market. Nineteen U.S. metro areas specialize in at least three of the five key industry subsectors.¹⁴⁵

Summary

The combination of world-leading research institutions, strong public and private investment, and a thriving domestic bioscience sector have ensured U.S. leadership in key subfields of biotechnology research—but the U.S. position is far from secure. These tailwinds may have propelled American biopower, but they require constant tending to endure.

American Biopower: Headwinds

The creation of the NSCEB reflects a recognition that biotechnology is essential to 21st-century national competitiveness and security, but also that existing policy is inadequate to secure U.S. leadership. Although powerful tailwinds have propelled the U.S. biotechnology leadership thus far, securing America’s position as the global biopower will also require navigating domestic headwinds, specifically:

1. **Inconsistent federal definitions, standards, and regulations**
2. **Insufficient public and private investment, especially for biotechnology applications beyond health and medicine**
3. **Inadequate biotalent**
4. **Insufficient and often inaccessible biodata due to regulatory ambiguities, underdeveloped infrastructure, and lack of a coordinated effort to maximize existing and future U.S. biodata**
5. **Lack of broad and durable public trust for sensitive next-generation biotechnology applications such as genomic editing and human-computer interfaces**

Definitions, Standards, and Regulations

The federal government lacks a consensus definition for biotechnology and the bioeconomy, which means Washington will struggle to track and improve what it cannot uniformly measure. Providing this consistent measure, however, is especially hard for the bioeconomy given the growing number of sectors it includes, from pharmaceuticals to energy, agriculture, textiles, and defense. Nevertheless, developing a single working definition of biotechnology and the bioeconomy is essential to better measuring progress and mobilizing investment. Today, different federal agencies maintain definitions of biotechnology relevant to their jurisdiction in areas such as agriculture, defense, and health.¹⁴⁶ None of the definitions capture the full spectrum of biotechnology techniques and applications. Compounding the challenge, key partners, allies, and international bodies have their own definitions of biotechnology, impeding global efforts to measure, track, and partner in the sector. Domestically, the lack of uniform definitions and terms confuses policy and undermines investment. Some investors reportedly view Europe's bioindustry as more appealing because the European Union (EU) has adopted uniform standards, unlike the United States.¹⁴⁷

The Biden administration recognized this challenge and advanced two solutions.¹⁴⁸ The first is to modify the North American Industry Classification System (NAICS) and the North American Product Classification System (NAPCS). These two systems generate federal codes to measure and track economic activity across different sectors and products. This provides a clear, uniform measure for government and third-party researchers to identify subtrends across the economy. However, this existing framework focuses on specific sectors and final products, whereas the bioeconomy cuts across multiple sectors and hinges more on the techniques used to create a product, rather than the product itself. In addition, a single company could use biotechnologies to generate multiple products across different sectors. The second option is for the U.S. Bureau of Economic Analysis (BEA) to create a satellite account, which provides more in-depth analysis of parts of the economy, such as outdoor recreation or tourism, that can be obscured by the BEA's high-level statistics.¹⁴⁹ However, a March 2023 report from the BEA concluded that, while developing a satellite account for the bioeconomy was technically feasible, refining it to capture specific parts of the bioeconomy was not due to insufficient data and a lack of consensus definitions.¹⁵⁰

In the Biotechnology EO, the Biden administration advanced its own definition of the bioeconomy as "economic activity derived from biotechnology and biomanufacturing."¹⁵¹ It also tasked the National Institute of Standards and Technology (NIST) with developing the Bioeconomy Lexicon, which itself advanced a broader definition of the bioeconomy as "economic activity derived from the life sciences, particularly in the areas of biotechnology and biomanufacturing, including industries, products, services, and the workforce."¹⁵² The lexicon seeks to harmonize terms to help measure and grow the bioeconomy, and NIST acknowledges that its definitions are subject to change given the rapidly evolving nature of the field. Although the administration rightly prioritized developing consensus definitions—and it has made progress through the BEA report and NIST lexicon—that consensus has yet to form.

The lack of consensus definitions and standards reflects a broader lack of a coherent, streamlined regulatory framework for biotechnology. Overlapping, burdensome, and often ambiguous federal regulations—primarily from the EPA, USDA, and U.S. Department of Health and Human Services (HHS)—continue to overload small and medium biotechnology enterprises with inconsistent definitions, standards, and approval timelines, despite repeated efforts across several administrations to make progress through the Coordinated Framework. Moreover, existing biotechnology regulations do not always apply to all biotechnology research, complicating efforts for the government to support and coordinate federal biotechnology research, while burdening researchers and innovators with a complex compliance regime.

Public Investment

Although federal investment in biotechnology R&D has been invaluable, it remains inadequate and imbalanced. Specifically, the United States must scale public and private investment in biotechnology ventures beyond health and medicine, as important as they are. The DoD's investments in BioMADE and DBMIP are vital but insufficient to build a truly world-class domestic bioindustrial base for the country's defense and commercial needs.

The leading U.S. government funders of biotechnology R&D are NSF and NIH. Despite considerable strengths, as structured, NSF and NIH face restrictions on the types of projects and applicants they can fund. NSF funds mostly fundamental research and education, and it tends to spread funding across its 10 biotechnology-relevant

directorates that manage a host of programs, which can dilute impact. America's Seed Fund—an NSF initiative that provides \$200 million annually to support technology prototyping and commercialization—shows promise, but supports a range of ventures, again diluting its benefit to the bioeconomy. For its part, NIH focuses on health and biomedical-related research. Although NSF and NIH provide vital support for early-stage research; STEM workforce development; and cross-sectoral collaboration, these restrictions limit their ability to catalyze the broader biotechnology sector.

Private Investment

Private investment—VC funding in particular—plays an indispensable role in seeding and scaling the U.S. biotechnology sector. However, the reliance on private funding exposes the sector to volatile market conditions and investor sentiment. The U.S. biotechnology sector experienced this firsthand following the pandemic. After peaking in 2021, private funding, IPOs, and mergers and acquisitions (M&A) activity for biotechnology collapsed, leading to layoffs, product delays, and even closures.¹⁵³ In 2022, the biotech IPO market collapsed 93 percent

The United States must scale public and private investment in biotechnology ventures beyond health and medicine.

from the prior year, and around 30 percent of public biotech companies in the United States and the EU had less than a year's worth of cash on hand.¹⁵⁴ More than half of the smaller, publicly traded biotech companies have insufficient cash to make it through 2025.¹⁵⁵

Compared to 2020 and 2021, there was a 54 percent decline in capital available to the biotech sector in 2022.¹⁵⁶ Many firms are still reeling from high-profile bank failures, notably the collapse of Silicon Valley Bank. The collapse in IPOs, declining access to capital, and greater risk aversion from investors has especially hurt smaller biotech companies. According to analysis from Ernst & Young, out of 223 companies that went public in 2020 and 2021 that remained publicly traded at the end of 2022, more than 90 percent saw lower market values compared to their IPOs, with an average decline in market value of more than 50 percent. The cash-constrained environment has led smaller firms to reduce spending, undercutting R&D.¹⁵⁷ Changing investor

Even as private investment recovers, investors remain sensitive to inherent risks in the biotechnology sector.

sentiment and higher interest rates also created more cautious capital, leading investors to opt for more established companies and applications.

In 2023, total VC and private equity investment in biotechnology was \$18.1 billion—32 percent less than the \$25.3 billion invested in 2022, and far short of the \$48.25 billion invested in 2021 and \$29.6 billion in 2020.¹⁵⁸ But the tide appears to be turning. In January 2024, biotechnology companies raised more than \$6 billion—the most since the pandemic-era boom—likely due to improved investor sentiment, a general market rebound, increased M&A activity across the sector, and expectations of lower interest rates.¹⁵⁹ As of November 2024, the closely watched SPDR Biotech exchange-traded fund (ETF), which shed almost two-thirds of its value since 2021, has recovered by more than 50 percent from its October 2023 low.¹⁶⁰

Even as private investment recovers, investors remain sensitive to inherent risks in the biotechnology sector, which include long development timelines, uncertain outcomes of clinical trials and commercial viability, and considerable regulation. Compounding investor reluctance are higher interest rates and a perceived boom and bust cycle for biotechnology companies such as Moderna, which saw demand for its products spike during the pandemic only to crater in its wake. This has especially limited available capital for early-stage biotechnology ventures, which rely on equity financing for initial, pre-market R&D. Early-stage biotechnology firms now face increased competition for initial capital, which could limit their R&D and increase pressure to demonstrate product viability earlier in development. In short, investors continue to see significant potential returns from biotechnologies but are hedging toward more proven firms and applications.

Rebounding private investment, IPOs, and M&A activity, however, have not changed the market's overwhelming preference for biotechnology applications in medicine, health, and pharmaceuticals. Between 2018 and 2021, less than 1 percent of all VC funding went to biofuels and agricultural chemicals, with virtually all the remaining capital going to biotechnologies related to medicine, health, and drug discovery and development.¹⁶¹ Maturing a broader U.S. bioeconomy will require public and private investment beyond traditional areas of focus in medicine, health, and pharmaceuticals. Trends in biotechnology patents mirror this dynamic. Of all the

patents in bioscience between 2018 and 2021, patents in agricultural chemicals comprised just 1.6 percent, plants 4.2 percent, and biopolymers 0.1 percent, compared to 66.1 percent for medical and surgical devices and pharmaceuticals.¹⁶²

The overwhelming focus of public and private funding on health and pharmaceutical applications of biotechnology has also developed a strong disposition among life scientists to view biotechnology through these lenses, which can limit demand signals for commercial and general research in applications in other areas such as energy, agriculture, material design, and manufacturing. Broadening interest from investors, scientists, talent, and policymakers in the broader biotechnology ecosystem is essential to securing U.S. leadership in a field far broader than health, medicine, and pharmaceuticals.

Biotalent

As with many other critical and emerging technologies, insufficient talent is an increasing headwind for American biopower. The wide range of skills and occupations within biotechnology reflects the sector’s breadth, as well as its rapid evolution and growth. Key sectors within the bioeconomy include the biomedical, bioindustrial, and agricultural sectors, each with their own permutation of occupations, skills, and training. The lack of a consensus definition for biotechnology, however,

makes it difficult to measure the sector’s total and projected employment, or employment demand. To be clear, U.S. biotechnology leadership does not require producing more biotechnologists per se, but rather combinations of biochemists, biophysicists, biological technicians, material scientists, laboratory technicians, chemical engineers, computer scientists, and technicians working together on specific products and applications. Efforts to improve STEM talent broadly will, by extension, alleviate skill shortages within biotechnology. According to the President’s Council of Advisors on Science and Technology (PCAST), the United States must increase the total number of undergraduates receiving STEM degrees by 34 percent every year to retain the country’s edge in science and technology.¹⁶³ At the same time, 90 percent of technology industry leaders say recruiting is a “major or moderate challenge.”¹⁶⁴

The domestic STEM workforce has always relied in part on robust and stable federal funding for general R&D. Domestic biotalent has depended specifically on general R&D in fields such as biological science, engineering, and information and computer sciences. Concerningly, a 2020 report from the NASEM found that “insufficient federal funding for U.S. universities and bioeconomy training programs has the potential to diminish the ability to produce and retain a skilled technical workforce.”¹⁶⁵ Two years later, the PCAST assessed that an insufficient workforce was inhibiting U.S.

Careers in the Future Bioeconomy Workforce

Biomedical	Agricultural	Bioindustrial
Scientist	Plant and animal scientist and geneticist	Biochemist
Biomedical engineer	Food scientist	Biofuels technology and product
Biomedical technician	Soil and plant scientist	development manager
Process engineer	Field technician	Fermentation engineer
Microbiologist	Laboratory assistant	Research engineer
Medical manufacturing technician	Fermentation engineer	Operations manager
Fermentation engineer		
Cross-Cutting		
	Computer systems analyst	
	Software developer	
	Data scientist	
	Machine learning engineer	
	Patent lawyer	
	Bioethicist	
	Computational genomics researcher	

Source: “Building the Bioworkforce of the Future,” The White House, June 2023, <https://www.whitehouse.gov/wp-content/uploads/2023/06/Building-the-Bioworkforce-of-the-Future.pdf>.

biomanufacturing capacity, limiting the ability to “meet the demand to scale up new bioproducts and the biomanufacturing infrastructure necessary to move products to pilot scale production. This has led some entrepreneurs and companies to move to Europe or Asia to begin scaling up production.”¹⁶⁶

There are also challenges at the postgraduate level. The NSF has tracked U.S. recipients of doctoral degrees since 1992. The following breakdown of biotechnology-related fields suggests that, while doctorates in key areas such as life sciences and biological and biomedical sciences have increased substantially, other important fields such as chemistry, agricultural sciences and natural resources, materials science engineering, and industrial and manufacturing engineering have seen only modest increases. Those fields remain, in total volume, significantly smaller.

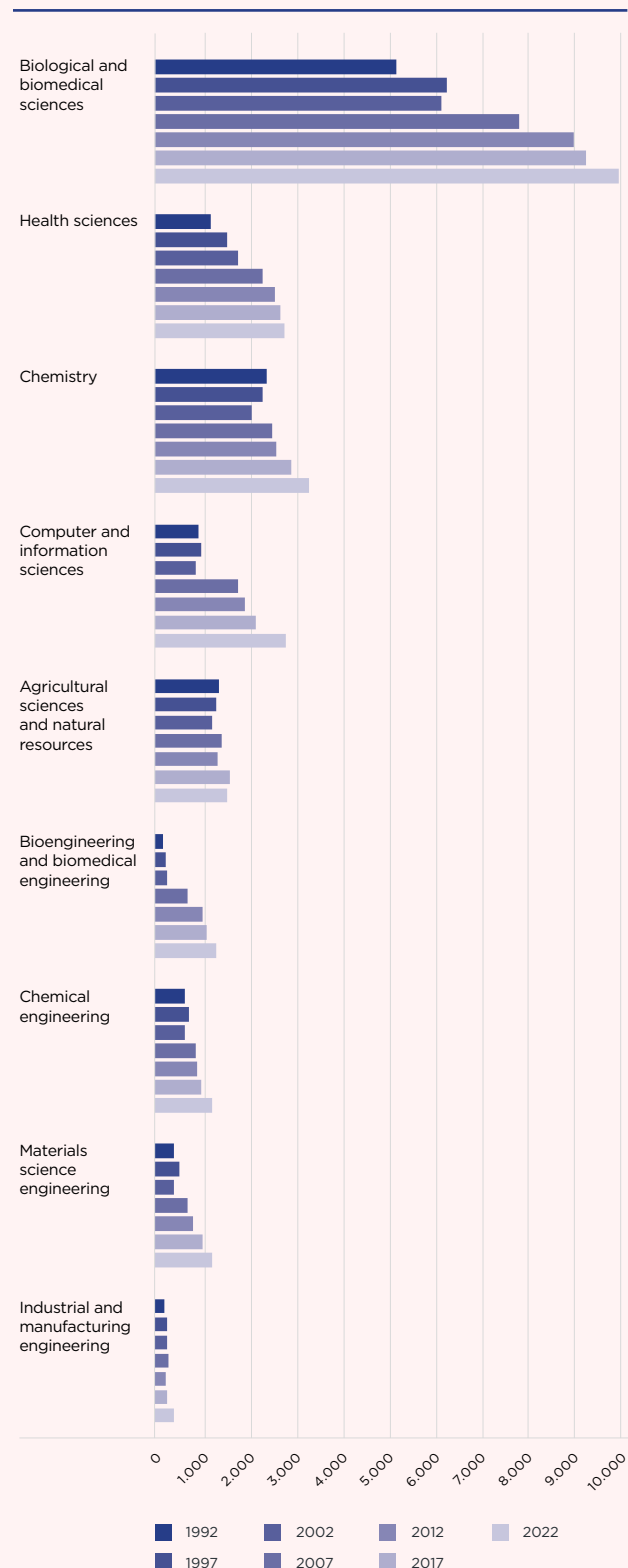
Data from the OECD, which has a slightly different method for organizing doctorate or equivalent degrees, suggests that U.S. graduates in biotechnology-related fields have remained largely stagnant since 2013.

The challenges to growing America’s biotalent coincide with several reports confirming rapid growth in the U.S. biotechnology sector and increasing demand for biotechnology-relevant skills. Data from the U.S. Bureau of Labor Statistics indicate that many biotechnology-linked occupations are projected to grow “faster” to “much faster” than other jobs in the U.S. economy more broadly.¹⁶⁷

Although doctorate recipients will continue to play a central role in driving biotechnology research and innovation, the bioeconomy has the potential to significantly expand employment in several occupations that require less than a four-year degree, let alone a PhD, including food science technicians, biological technicians, and manufacturing product technicians. Indeed, some biotechnology employers have reportedly hired community college students before they even graduate.¹⁶⁸

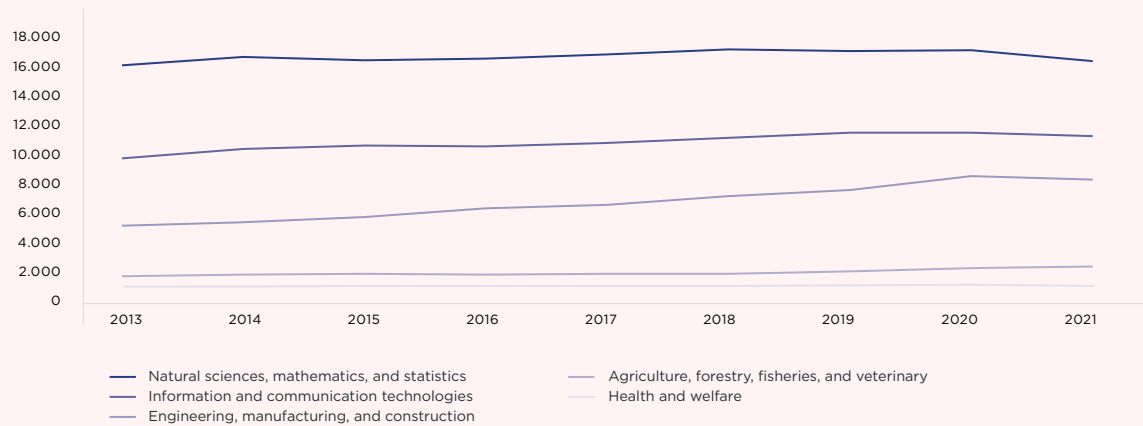
Growing private sector demand for biotechnology professionals, however, could undermine the training and innovation pipeline by luring graduates away from postgraduate academic opportunities by offering much higher salaries. Between 2021 and 2022, there was a 10 percent drop in the number of U.S. citizens and permanent residents pursuing postdocs in biological and biomedical sciences.¹⁶⁹ Similarly, survey data from NSF indicates that fewer U.S. doctoral recipients than ever before are pursuing postdocs in the life sciences.¹⁷⁰ Robust federal funding for general R&D in life sciences can mitigate the hemorrhaging of PhDs to industry and ensure that today’s private biotechnology demands do not inadvertently stifle tomorrow’s biotechnology workforce.

Uneven Growth in U.S. Doctorate Recipients in Key Biotechnology Fields



Source: “Doctorate Recipients from U.S. Universities: 2022,” U.S. National Science Foundation, September 28, 2023, <https://nces.nsf.gov/pubs/nsf24300>.

Stagnation in U.S. Doctorate (or Equivalent) Recipients Across Biotechnology Sectors



Source: "OECD Data Explorer," Organisation for Economic Co-Operation and Development, accessed September 20, 2024, <https://data-explorer.oecd.org>.

Biotechnology Job Growth Outpaces the Broader Economy

U.S. Occupation	Employment Growth Rate	Growth Rate Relative to All U.S. Occupations
All U.S. occupations	3%	Average
Biochemists and biophysicists	7%	Faster
Bioengineers and biomedical engineers	5%	Faster
Biological technicians	5%	Faster
Chemical and material scientists	6%	Faster
Chemical engineers	8%	Faster
Computer and information research scientists	23%	Much Faster
Clinical laboratory technologists and technicians	3%	Faster
Database administrators and architects	8%	Faster

Source: "Occupational Outlook Handbook," U.S. Bureau of Labor Statistics, August 29, 2024, <https://www.bls.gov/ooh>.

A closer look at doctoral recipients in biotechnology-related fields reveals a broader headwind to U.S. biotechnology competitiveness and U.S. technology competitiveness generally: the nation's broken immigration system. In 2022, temporary visa holders earned 64 percent of doctorates in computer and information sciences, 57 percent of doctorates in engineering, and 25 percent of doctorates in biological and biomedical sciences. In 2015, across biological, agricultural, and environmental sciences, foreign students represented 15 percent of the workforce with a bachelor's degree, 27 percent of workers with a master's degree, and 47

percent of the workforce with a PhD (this includes naturalized immigrants and permanent residents, not just visa holders). The last time the U.S. Congress meaningfully reformed its immigration system was in 1990. Country caps—a policy dating to the 1960s—suffocate America's access to high-tech talent and have fed a backlog of 1.4 million people waiting for green cards.¹⁷¹ It would be a mistake to take U.S. high-tech immigration for granted. As demand for biotechnology grows abroad, along with strong inducements from talent-hungry governments, so will opportunities for top foreign talent to remain abroad.

To unleash the U.S. biotechnology sector, it will take more ambitious efforts across government, academia, and the private sector to foster and fund biotechnology-relevant skills, training, and careers to match expected demand.

Biodata

The United States will struggle to maintain its biotechnology leadership without reforms to unlock the potential of its biodata. Data is perhaps the United States' biggest untapped biotechnology asset, comparable to the nation sitting on a massive gold mine it is blocked from accessing by its own rules and regulations.¹⁷²

Given the breadth of the bioeconomy, driving research and innovation requires access to a broad range of data. This includes conventional biodata that describes the genomes, nucleic acid transcripts, proteins, and metabolites of living organisms from humans to plants to microbes.¹⁷³ It also includes nutrition, agricultural, environmental, satellite, and industrial data, which can combine with conventional biodata to surface insights about new medical treatments, pharmaceuticals, energy sources, climate-resistant crops, and novel biomanufacturing processes. The greatest biotechnology insights will emerge from analysis of diverse datasets.¹⁷⁴

The challenge for the United States, however, is both a lack of biotechnology-relevant data and integrated, structured, and secure access to that data for research, development, and innovation. The United States continues to amass significant biodata and associated metadata from a range of sources, including government agencies and a combination of federally funded and private research-

commercial value of their underlying data has encouraged further hoarding and consolidation.¹⁷⁸

The federal government has also seized on the collapsing cost of gene sequencing to drive several efforts to securely amass biodata and make it available. NIH's All of Us Research Program seeks to build a diverse, voluntary database of Americans' health information for internal and legitimate third-party research. So far, the program has made more than 287,000 EHRs and 245,000 whole genome sequences available for researchers.¹⁷⁹ The All of Us database is high quality and growing, but it remains relatively small. NIH also houses the Genomic Data Commons, which provides cancer researchers with a unified repository of "standardized cancer genomic data sets."¹⁸⁰ Another major effort is the Department of Veterans Affairs' Million Veteran Program, which has collected voluntary health and genetic information from a million veterans for research to improve veterans' health and support new treatments.¹⁸¹ It is now one of the world's largest research programs for studying the relationship between genes and human health.¹⁸² The USDA's Agricultural Research Service (ARS) also boasts "the world's largest collection of agricultural information," including significant genetic data on plants and animals through its National Agricultural Library.¹⁸³

The federal government has taken some steps to increase publicly available biodata. This began with the Office of Science and Technology Policy's (OSTP's) 2013 Public Access Memorandum and continued with its 2022 memorandum on "Ensuring Free, Immediate, and Equitable Access to Federally Funded Research," which outlines steps to make federally supported scientific data, including biodata, publicly available.¹⁸⁴

Still, biotechnology-relevant data today is generally not findable, accessible, interoperable, and reusable (FAIR).¹⁸⁵ The proliferation of biodata, however, has created an increasingly complex and siloed data landscape as researchers and developers follow their own practices for data collection, labeling, storage, access, and security. In some cases, missing metadata raises quality concerns. In others, misaligned standards for labeling, access, or security constrain researchers' abilities to share, integrate, validate, and use different data types. Cleaning and curating data from disparate sources is also costly and time intensive, creating another barrier to research, development, and innovation. Despite these challenges, there remains no coordinated national effort to develop interoperable tools or uniform standards for biodata. Efforts to integrate and even consolidate datasets, however, must also guard against heightened cybersecurity risks from data poisoning, reidentification, and exfiltration.

Data is perhaps the United States' biggest untapped biotechnology asset.

ers.¹⁷⁵ Increasingly advanced tools and research methods have also increased the complexity of available biodata. The collapsing cost of gene sequencing in particular has led to a proliferation of epigenomic data by companies such as 23andMe and Ancestry, which now have the gene sequences of an estimated 26 million people worldwide.¹⁷⁶

Another driver of proliferating biodata is the transition among health providers to electronic health records (EHRs), which now number more than 250 million.¹⁷⁷ However, EHRs are typically managed by private software firms, which have a strong profit incentive to amass data. Indeed, expectations by these firms that advancements in AI and biotechnology will substantially increase the

At the same time, it is not enough to simply amass biodata. The data must be securely stored, structured, labeled, secured, and analyzed, which requires significant resources for digital infrastructure, computational resources, trained personnel, and cybersecurity.¹⁸⁶ The required resources will only grow as relevant data becomes more abundant and heterogeneous over time.

The United States must also find a way to unleash its biodata without compromising its commitment to privacy and civil rights.

All of this points to significant headwinds for the United States' abilities to identify, develop, and commercialize biotechnology discoveries. The fragmented, disorganized landscape of U.S. biodata is not the only challenge. The United States must also find a way to unleash its biodata without compromising its commitment to privacy and civil rights. This necessitates a strategy for using *deidentified* health and genomic data, in which personally identifiable markers of the data have been removed. But herein lies a major challenge: there is considerable ambiguity about who owns deidentified patient data in the United States.¹⁸⁷

HIPAA is generally understood to give patients ownership of their medical data. However, it has less to say about who owns that data once it has been deidentified, or how companies should be compensated for using it.¹⁸⁸ This ambiguity has incentivized data hoarding by private firms, which remain unsure about if and how they can sell their data, which in turn undercuts data sharing and life sciences research.

Another challenge for U.S. biodata is retention. Valuable biodata is often deleted after several years as health providers comply with various state laws that require deletion or compliance with HIPAA's requirements for appropriate data safeguards. Generally, AI tools can extract more value from more data, so the deletion of health data over time can undercut future insights and innovation. At the same time, publicly funded databases such as NIH's National Center for Biotechnology Information (NCBI) are not structured to absorb the full range of biodata, which has grown both in volume and complexity as increasingly powerful sequencing platforms emerge. The United States' failure to develop and fund permanent infrastructure that can handle the nation's growing biodata results in the needless deletion or siloing of this critical asset.

Finally, the United States must balance its historic commitment to open data—including biodata—with risks of exploitation by China. The United States has promoted open, scientific research going back more than 200 years to the founding of the National Library of Medicine. This commitment is foundational to NIH, which has long made scientific information accessible to researchers through publicly accessible databases. The United States' commitment to open science has accelerated research and discovery while reinforcing scientific norms around openness, transparency, and accessibility around the world. But some competitors such as China never fully embraced that model. Although China has amassed global biodata through open sources and other licit and illicit means, it has not made its own biodata publicly available, even as it pursues a well-resourced, coordinated effort to collect and organize its biodata assets in a bid for global biotechnology dominance. The February 2024 Executive Order on Preventing Access to Americans' Bulk Sensitive Personal Data and United States Government-Related Data by Countries of Concern restricted China's access to American biodata and was an overdue and welcome step.¹⁸⁹ Still, absent further efforts to better organize and unleash U.S. biodata, China will likely be in a stronger position to apply advanced AI tools to its existing biodata resources, with the potential to supercharge its biotechnology research and innovation. In the age of AI, nations with large, structured, accessible, and secure life science databases will enjoy a significant and potentially decisive advantage in biotechnology.

To its credit, the Biden administration has made important progress in elevating the importance of data in driving U.S. biotechnology leadership. In December 2023, the Interagency Working Group on Data for the Bioeconomy and the National Science and Technology Council released a report identifying many of the key problems described above. The report made a strong

In the age of AI, nations with large, structured, accessible, and secure life science databases will enjoy a significant and potentially decisive advantage in biotechnology.

case for making “data FAIR across domains, removing unnecessary siloes that limit data sharing, integration, and utilization.”¹⁹⁰ It also called for an “interoperable, networked data fabric” supported by robust, stable

federal funding for the digital infrastructure, computational resources, and trained personnel required to make the most of government-owned and federally funded biodata.¹⁹¹

Public Trust

A final but less understood headwind to U.S. biotechnology leadership is eroding public trust and growing partisan polarization around scientific expertise. Analysis of the 2022 General Social Survey by the Associated Press-NORC Center for Public Affairs Research shows that the percentage of adults with “a great deal of confidence” in the scientific community fell from 48 percent in 2018 to just 39 percent in 2022.¹⁹² Politically charged debates over the extent of pandemic-era restrictions likely fueled this trend: in 2018, 45 percent of Republicans expressed a great deal of confidence in the scientific community, but by 2022, the number had fallen to 22 percent.¹⁹³ The same year, the share of Republicans who expressed a great deal of confidence in medicine fell to just 26 percent.¹⁹⁴ A May 2023 survey showed that just 69 percent of Americans have trust in scientists to act in the public’s best interest, declining from 86 percent in January 2019 (trust in medical scientists declined from 89 percent to 75 percent over the same period).¹⁹⁵ Partisan polarization around vaccines has been especially acute. Among Republicans, 67 percent have concerns about serious adverse effects of vaccines, compared to only 27 percent of Democrats.¹⁹⁶

Although partisan polarization around COVID-19 vaccines offers a more recent example of public backlash to biotechnologies, genetically modified organisms (GMOs) offer another cautionary tale. Despite global scientific consensus about the safety and value of GMOs, considerable public skepticism remains nearly three

decades after GMOs became broadly available for public consumption. A 2015 survey from the Pew Research Center found that, while 88 percent of scientists thought genetically modified (GM) foods were safe to eat, just 37 percent of the public agreed.¹⁹⁷ The public’s views seemed to worsen over time, with a 2020 Pew survey finding that just 27 percent of Americans viewed GM foods as safe to eat. Skepticism of GM foods is global; across the 20 countries surveyed by Pew, a median of just 13 percent said GM foods were safe, compared to 48 percent who viewed them as unsafe.¹⁹⁸

If the American public remains skeptical about GM foods three decades after their introduction, it is reasonable to assume significant public skepticism of more interventionist, science fiction–like biotechnologies such as human gene editing and human-machine interfaces. Strong and sustained public investment in biotechnologies will require, at a minimum, the absence of intense public opposition, and ideally, robust and stable public support. If powerful commercial interests in sectors such as agriculture, defense, energy, mining, plastics, textiles, or heavy industry view biotechnologies as potential threats to their business model—instead of as market opportunities—it could engender organized opposition that risks stifling public investment and policy.

Summary

The United States occupies one of the strongest positions to lead the biorevolution. Its dynamic capital markets, innovative private sector, world-leading higher education institutions, and unrivaled STEM talent bestow tremendous advantages. To secure U.S. biotechnology leadership, however, Washington must act with urgency to intensify tailwinds and diminish headwinds. In the meantime, U.S. competitors race to close the gap.

Chapter 03

Biopower Abroad

Countries worldwide have awoken to the promise of biotechnologies, and many are pursuing ambitious strategies to secure their places in the biorevolution. Securing America's place as the global biopower requires a clear-eyed assessment of its competition abroad in a fast-changing landscape. Strengthening U.S. biotechnology leadership also demands an appreciation for its global context: biological research, assets, supply chains, capital, and human talent often cross national jurisdictions, along with biotechnology-enabled dangers such as novel pathogens, GM invasive species, and even the export of totalitarian biosurveillance to enable social control.

This section surveys the emerging global architecture for biotechnology—including bilateral, plurilateral, and regional initiatives—and analyzes the strengths and weaknesses of key biopowers abroad, including the EU, Belgium, France, Switzerland, the UK, Japan, South Korea, and China.

Global Architecture

The global architecture for biotechnology—the international and multilateral laws, policies, and norms—remains embryonic. The domestic struggle to define and govern biotechnology and the bioeconomy compounds at the international level, with different nations adopting definitions that reflect their biotechnology ambitions, priorities, and concerns. Although this divergence is natural given the sector's breadth and relatively early stage of development, some convergence and consensus around definitions, terms,

metrics, and norms is necessary to begin constructing a global biotechnology architecture.

The divergence begins with definitions. For instance, the term “bioeconomy” features prominently in strategic plans for Brazil, China, Germany, the EU, South Africa, and the United States. By contrast, Australia, Belgium, and New Zealand reference the bioeconomy primarily as a means to achieve sustainability objectives.¹⁹⁹ Other technology-leading nations such as Canada, India, Japan, Singapore, South Korea, and the UK do not define the bioeconomy in major policy documents.²⁰⁰ The terminological inconsistency in national policies reflects the lack of global consensus about both the definition and the strategic importance of biotechnology and the bioeconomy.

At the international level, there is no comprehensive governance regime for biotechnology. The first major, related international agreement was the 1972 Biological Weapons Convention (BWC), which effectively prohibited the development, acquisition, and use of biological weapons.²⁰¹ At the same time, the BWC acknowledged the value of biological research for peaceful purposes in areas such as agriculture and medicine, and it encouraged international cooperation in these domains. Although the BWC established a robust and broadly stable international norm against biological weapons, it continues to lack a binding verification mechanism—a weakness underscored by the exposure of biological weapons programs in the Soviet Union, Iraq, and North Korea. The agreement also failed to address the threat from nonstate actors who could employ dual-use technology to make bioweapons.

Two decades later, the international community passed another significant agreement related to biotechnology: the 1992 United Nations Convention on Biological Diversity (CBD).²⁰² The Convention focused on conserving biological diversity and ensuring the sustainability of biological resources, including genetic resources. To that end, it established a global framework to promote fair and equitable access to and use of genetic resources. Disagreement about whether signatories have fulfilled this commitment remains a source of international contention.

In 2000, the international community supplemented the CBD with the Cartagena Protocol, which regulates the transnational movement of organisms modified by modern biotechnologies, such as animals and plants, that may harm biological diversity and human health.²⁰³ A decade later, the international community again amended the CBD with the Nagoya Protocol, which established a mechanism for the fair and equitable sharing of benefits from the use of genetic resources.²⁰⁴ The CBD and its subsequent protocols helped facilitate a global market for biotechnology products by establishing a framework that emphasized safety, public input, national consent, and information sharing.

To date, the international community's only agreements governing bioethics are the nonbinding Universal Declaration on the Human Genome and Human Rights and the International Declaration on Human Genetic Data, which the United Nations Educational, Scientific and Cultural Organization (UNESCO) adopted in 1997 and 2003, respectively.²⁰⁵ The 1997 declaration provided the first universal framework for ethical human genetic research, clinical practices, and policymaking. Notably, it emphasized that research must "fully respect human dignity" and prohibited discrimination based on genetic characteristics and the reproductive cloning of humans. The 2003 declaration reemphasized many of these principles and promoted norms around international collaboration, responsible genetic research and genetic data handling, and equitable access to the benefits of genetics-related research. Both declarations remain cornerstones of the emerging global architecture for responsible genetic research. Although it is nonbinding, the World Health Organization (WHO) released a framework in 2022 for the responsible use of life sciences, which included recommendations for member states to limit biorisks and govern dual-use research.²⁰⁶

Limited activity at the international level has not stopped organic, global efforts to promote biotechnology cooperation, research, and innovation. For instance,

the Global Alliance for Genomics & Health (GA4GH) is an international nonprofit that convenes leading health care, research, ethics, government, life sciences, and technology organizations to promote consensus around standards, policy frameworks, and tools for genomic medicine.²⁰⁷ Another international nonprofit, the International Genetically Engineered Machine (iGEM), helps cultivate global talent and norms in synthetic biology through an annual competition in which students from around the world develop and demonstrate applications of synthetic biology.²⁰⁸ International declarations and initiatives such as iGEM and GA4GH may not be legally binding, but they have helped to raise global awareness and cooperation about biotechnology applications and to reinforce norms around responsible research and use.

Bilateral, Plurilateral, and Regional Initiatives

Bilateral, plurilateral, and regional initiatives have also formed as part of an emerging global architecture for biotechnology. In 2017, Brazil, Russia, India, China, and South Africa (BRICS) launched the BRICS Working Group on Biotechnology and Biomedicine to promote collaboration on biotechnology R&D.²⁰⁹ The Association of Southeast Asian Nations (ASEAN) first established a Working Group on Biotechnology in 1983 and has recently focused on biotechnology applications for agriculture, convening a workshop in Jakarta, Indonesia, in March 2024 with support from the U.S. government.²¹⁰

At the bilateral level, no major agreements focus exclusively on biotechnology. Instead, biotechnology has fallen within broader bilateral agreements concerning R&D, trade, and technology cooperation. Australia and Singapore, for instance, concluded a Digital Economy Agreement in 2020 that established a framework for digital trade to promote cross-border dataflow, reduce barriers to digital trade, strengthen intellectual property (IP) protection, and promote standards harmonization—all of which are essential to stronger biotechnology-related cooperation.²¹¹

Transatlantic biotechnology cooperation stretches back to 1990 with the creation of the EC-U.S. Task Force on Biotechnology Research between the European Commission (EC) and the United States, which provided a forum for coordination and discussion between European and U.S. researchers on biotechnology applications for the public benefit.²¹² The U.S.-EU Trade and Technology Council (TTC), launched in 2021, advances

transatlantic governance of critical and emerging technologies rooted in shared democratic values. Although one of the TTC's working groups on technology standards cooperation included biotechnology, the TTC did not explicitly reference biotechnology cooperation and promoting the bioeconomy as a priority until its sixth ministerial meeting in April 2024.²¹³

Transatlantic biotechnology cooperation has often encountered differences in regulatory, policy, and normative barriers concerning GMOs. The United States has adopted a more market-driven approach, with producers widely adopting GM crops and livestock, while the EU maintains far stricter regulations that reflect considerable public concerns about the alleged health and environmental impacts of GMOs. Proactively anticipating similar divergences for future biotechnology applications—especially for any GM products and services in agriculture and health—could minimize transatlantic regulatory gaps and points of friction that could impede cooperation.

Multilateral security partnerships have also increased their focus on biotechnology as part of a broader emphasis on critical and emerging technologies. NATO first identified biotechnologies and human enhancement (BHE) technologies as a priority technology area in 2019 and released the first international strategy to govern BHE development and use in April 2024.²¹⁴ The strategy calls for enhanced biotechnology investment and collaboration to improve production of defense-relevant materials, inputs, and capabilities. Thus far, concrete efforts appear limited to developing risk and impact assessment toolkits, promoting education and awareness of BHEs, and fast-tracking select BHE technologies.

In 2021, NATO launched the Defence Innovation Accelerator for the North Atlantic (DIANA) to run competitive challenges for industry to develop “deep tech dual-use technologies” for critical defense and security problems.²¹⁵ The next year, the alliance also launched the NATO Innovation Fund (NIF), a more than \$1 billion, multisovereign VC fund to make long-term investments in start-ups developing defense and security-related technologies.²¹⁶ Both initiatives have prioritized biotechnologies.²¹⁷

In September 2021, the United States, the United Kingdom, and Australia announced a trilateral security partnership called AUKUS.²¹⁸ Pillar II of the partnership calls for enhancing joint capabilities and interoperability in key emerging technologies. Although the initial Joint Leaders Statement did not single out biotechnology as a focus, it remains ripe for cooperation—especially given the biotechnology leadership of the United States and the UK.

Another promising venue for biotechnology cooperation is the Quadrilateral Security Dialogue between the United States, India, Japan, and Australia, also known as the Quad. In May 2023, leaders of the Quad launched the Quad Investors Network (QUIN), an independent body to facilitate cooperation, innovation, and technological development across the four technology-leading nations.²¹⁹ Across the four governments, QUIN convenes investors, businesses, and governments to facilitate joint ventures and co-investments in strategically important technologies, including biotechnology.

The growing trilateral partnership between the United States, Japan, and the Republic of Korea is another ripe opportunity to strengthen joint biotechnology research, development, and investment. All three countries have a clear interest in strengthening their biotechnology sectors, and leaders should launch a trilateral biotechnology industry working group and negotiate a trilateral biotechnology cooperation agreement outlining opportunities for joint investment, trade, and talent exchange, as well as policy and regulatory harmonization.²²⁰

The trilateral partnership recently joined forces with the EU and India to launch the Biopharma Coalition (Bio-5) in June 2024 to boost biopharmaceutical supply chain resilience, with an initial focus on active pharmaceutical ingredients that are now predominantly sourced from China. The White House's announcement of Bio-5 also laid the groundwork for future coordination on policy, regulation, and R&D.²²¹

A Survey of Global Biopowers

To date, at least 49 countries have developed strategies related to the bioeconomy, but only a few have the combination of natural strengths and ambitious policies and investments to reach the higher tiers of national biopower.²²² Among them, only China stands as a serious rival to surpass the United States as the global biopower, although the United Kingdom and South Korea stand out for punching far above their weights, with France, Belgium, Switzerland, and Japan also making impressive progress. Each holds lessons for U.S. policymakers.

Many competing biopowers, including Japan and South Korea, rolled out detailed biotechnology strategies years ago. The UK has gone further, introducing a strategy in 2023 specifically for engineering biology. South Korea's recent strategy leans into specialization by focusing on “digital biotechnology” at the intersection of biotechnology, data, and AI. Most important, these nations have paired new strategies with new funding

for implementation. The EU's Bio-Based Industries Joint Undertaking (BBI JU), a public-private partnership to accelerate novel biobased solutions, came with a nearly \$4 billion mobilization.²²³ France's Innovation Santé 2030 initiative devoted nearly \$7.5 billion to boost health care innovation.²²⁴

Many competing biopowers, including Japan and South Korea, rolled out detailed biotechnology strategies years ago.

Funding is necessary but insufficient; biotechnology data and infrastructure also matter. The UK leads the world in responsibly consolidating biodata and making it accessible for research and innovation through the UK Biobank. The UK also has started to build cutting-edge R&D infrastructure, including laboratories, testbeds, and factories, to bridge the gap between research and business-critical technologies.²²⁵ Switzerland highlights the power of cultivating biotechnology “clusters” that concentrate research, industry, and capital, while Belgium underscores the importance of a favorable regulatory environment and public-private collaboration.

Biotechnology policy, investments, and activity abroad continue to grow, offering both examples and cautionary tales for U.S. policymakers. Below is a brief survey of leading global biopowers and their latest biotechnology developments and challenges.

The European Union

The EU has recognized the need to balance its historic emphasis on safety and regulation with biotechnology's potential for other priorities such as health, sustainability, climate change, and economic security. Biotechnology is one of the fastest-growing industries in the EU, contributing \$33 billion to its GDP and creating nearly 211,000 direct jobs across the economy in 2018.²²⁶ Between 2008 and 2018, the EU's biotechnology sector grew twice as fast as its economy.²²⁷ The EU also boasts a strong life science research base—with high-quality health, agricultural, and industrial biotechnology publications.

In 2012, the EU adopted a bioeconomy strategy that focused on addressing “the ecological, environmental, energy, food supply and natural resource challenges” Europe faces.²²⁸ The strategy featured a 12-step Bioeconomy Action Plan that called for, among other initiatives, greater investment, multidisciplinary research, and human capital; increased synergy between

bioeconomy-related policies; and standards development for biobased products to support new markets and help scale the European bioeconomy. A 2018 review of the strategy found that it had successfully mobilized research and investment funding, spurring shifts among member states to recognize the bioeconomy's potential and supporting new standards for biobased products.²²⁹ At the same time, it called for steps to further mobilize investments, ensure a stable regulatory environment, and promote policy coherence.

A key driver of the EU bioeconomy has been Horizon 2020, a roughly \$84 billion research and innovation fund that ran from 2014 to 2020.²³⁰ One of its principal objectives was promoting European science and R&D, and it funded various biotechnology-related projects.²³¹ The effort will continue through 2027 as Horizon Europe, which further emphasizes sustainability and climate change in alignment with the European Green Deal, which seeks to make the EU carbon neutral by 2050.

Arguably, Horizon 2020's most significant contribution to the European bioeconomy was the establishment of the BBI JU—a partnership between the EU and industry to accelerate innovation, development, and deployment of novel biobased solutions. Between 2014 and 2021, the BBI JU mobilized nearly \$4 billion.²³² The initiative rebranded as the Circular Bio-Based Europe Joint Undertaking (CBE JU) to emphasize the role of biobased products in promoting a sustainable, circular bioeconomy aligned with the European Green Deal.²³³ To date, the effort has supported several projects, from exploring the use of microorganisms and newly designed enzymes to “eat” otherwise nonbiodegradable plastics to an effort to use lignin—an abundant natural polymer found in most plants—to replace certain petroleum-based polymers used in textiles and other fabrics.²³⁴

In March 2024, the European Commission issued a communication regarding “Building the Future with Nature: Boosting Biotechnology and Biomanufacturing in the EU.”²³⁵ The document called for establishing an EU Biotech Hub by the end of 2024 to help researchers and businesses navigate regulations, new funding mechanisms to support EU biotechnology and biomanufacturing, and additional steps to streamline EU regulations. These steps could lay the groundwork for a future EU Biotech Act in the next few years, like the EU's recently adopted AI Act.²³⁶

The communication identified three headwinds to the EU's biotechnology leadership. First, EU biotechnology companies struggle to transition from lab research to commercialization, owing to the diffusion of biotechnology talent across member states and the dearth of world-leading biotechnology centers of excellence.

Second, it cited complex and lengthy permitting processes, and the patchwork of regulations between the member state and EU levels. For example, it takes up to three times as long to approve a biological plant protection product in the EU as it does in the United States. Finally, the communication noted challenges for biotechnology start-ups in accessing finance. The long-term, uncertain outlook of biotechnology firms is often at odds with the traditional, loan-based financing through banks more common in the EU. The EU's fragmented capital market also deprives later-stage firms of accessing the larger amounts of VC required to scale.

Although not explicitly tied to biotechnology, the EU's General Data Protection Regulation (GDPR) strictly regulates the processing of personal data, including genetic data, in biotechnologies.²³⁷ GDPR is a double-edged sword for the EU, as it provides a clear legal regime for the acquisition, use, and security of biodata and genetic data. This could in turn foster greater public trust that public and private researchers are using such data responsibly. On the other hand, the GDPR's strict rules for data processing, anonymization, and consent may limit valuable uses of biodata for R&D. By contrast, the lack of significant U.S. rules governing biodata—and the absence of a U.S. national privacy law generally—undermines cooperation with like-minded partners such as the EU, which has much stronger protections under the GDPR.

In short, the EU retains significant biotechnology strengths—a strong life science research base and educated workforce; clear policy ambitions for biotechnology concerning climate change, sustainability, and economic security; relative regulatory clarity for processing health and genetic data; and significant natural resources for a circular bioeconomy. However, its long-term biotechnology competitiveness will hinge on whether it can maintain these strengths while overcoming challenges related to the diffusion of capital, talent, policy, and regulation across member states.

The clear biotechnology leaders within Europe are Belgium, France, Switzerland, and the United Kingdom.



Biopower: The European Union

Tailwinds

- Strong life science research base
- Highly educated workforce
- Biotechnology ambitions linked to broadly popular goals such as economic security, sustainability, and fighting climate change

Headwinds

- Transition from lab research to commercialization
- Lack of world-leading biotechnology centers of excellence
- Lengthy permitting and complex, overlapping regulations
- Diffuse talent and capital markets across member states makes labor and financing a challenge for start-ups

Belgium

Belgium punches far above its weight for biotechnology, owing to a favorable regulatory environment, strong research infrastructure, and a culture of collaboration between government and industry.²³⁸ The country is home to more than 140 biotechnology companies, representing 7 percent of all such European companies.²³⁹ Belgium also accounts for almost 10 percent of Europe's biotechnology R&D spending, representing just 2.6 percent of the EU's total population.²⁴⁰ Belgium's biotechnology leadership benefits from a robust research infrastructure with more than 23 incubators, 19 research parks, and 8 universities, including biotechnology-forward institutions such as KU Leuven.²⁴¹ Belgium is also home to a robust biopharmaceutical industry—owing to its high-quality academic and research infrastructure—with more than 70 hospitals, nationwide research centers, and high-quality researchers.²⁴² A 2018 Belgium law also improved the regulatory environment by enabling rapid approval procedures for clinical trials, which can take as little as two weeks.²⁴³ Worldwide, Belgium ranks 9th for total patent filings per capita, propelled recently by a thriving biotechnology sector that saw a 28 percent increase in patent applications from 2022 to 2023.²⁴⁴



Biopower: Belgium

Tailwinds

- Favorable regulatory environment
- Strong research infrastructure and public R&D spending
- Culture of collaboration between government and industry

Headwinds

- Relatively small domestic market

France

France is a biotechnology leader, owing to longtime strength in health care innovation and robust government funding.²⁴⁵ The country is home to 820 biotechnology companies, including world-leading biopharmaceutical companies such as Sanofi at the vanguard of global vaccine development and deployment.²⁴⁶ Paris has emerged as a hub for European biotechnology start-ups, including DNA Script, one of the world's first companies to develop benchtop DNA printers.²⁴⁷ In 2021, the French government boosted its biotechnology sector through Innovation Santé 2030, which aims to reinvigorate and secure France's leadership in health care innovation with nearly \$7.5 billion in funding to boost biomedical research to accelerate new therapies and technologies, invest in priority areas including biotherapies and bioproduction, and streamline regulatory processes for clinical trials, among other steps.²⁴⁸ France is also home to several biotechnology research clusters and leading research institutions, from the renowned Institut Pasteur, which focuses on biomedical research, to Genopole, a biocluster south of Paris focused on genetics, genomics, and biotechnology.²⁴⁹ Despite substantial public investment, two-thirds of all French biotechnology companies cite access to capital as their top challenge, followed by hurdles in developing industrial partnerships and overcoming regulation.²⁵⁰



Biopower: France

Tailwinds

- Robust health and pharmaceutical sector
- Paris is an emerging hub for biotechnology start-ups
- Strong public investment through Innovation Santé 2030

Headwinds

- Underdeveloped industrial partnerships
- Access to capital
- Complex regulation

Switzerland

Like Belgium, Switzerland overperforms in biotechnology relative to its size. Switzerland's financial stability, robust international partnerships, strong scientific research industry, and pro-innovation culture propel this leadership. This dynamic benefits not only biotechnology but innovation broadly, explaining why the World Intellectual Property Organization has placed Switzerland at the top of its Global Innovation Index for the past 13 years.²⁵¹ Indeed, life sciences has been Switzerland's top export sector since 2013. The Swiss biotechnology sector has thrived under these conditions. Capital investments grew substantially during the pandemic and, in 2023, Switzerland's biotechnology sector posted record revenues of more than \$8 billion and raised nearly \$2.25 billion, a 50 percent increase over the previous year.²⁵² Switzerland is also in the top 10 nations worldwide for biotechnology patents, although its relative share is dwarfed by the United States, China, Japan, and South Korea.²⁵³ Biotech clusters, including the BioValley in Basel, the Biopole in Lausanne, and the Technopark in Zurich, have also driven Swiss biotechnology leadership by concentrating capital, life sciences research, and start-ups while fostering cultures of innovation and collaboration.²⁵⁴ Novartis and Roche are also world-leading biopharmaceutical companies, pushing the frontier in areas such as biotherapeutics and personalized medicine.²⁵⁵



Biopower: Switzerland

Tailwinds

- Robust international partnerships
- Strong scientific research base
- Pro-innovation culture and regulatory environment

Headwinds

- Intense competition for skilled labor and capital
- Dependence on international talent
- High-cost market

The United Kingdom

The UK ranks among the top global biopowers.²⁵⁶ It has a strong scientific and research tradition, with 2 of the top 10 universities in the world—Oxford and Cambridge.²⁵⁷ U.S. News & World Report ranks Cambridge as the fifth-best university for biotechnology and applied microbiology worldwide.²⁵⁸ In 2021, Cambridge's renowned Biomedical Campus alone added roughly \$2.8 billion to the UK economy.²⁵⁹

Together with London, the cities of Oxford and Cambridge form the so-called Golden Triangle for life sciences, concentrating world-class talent, research facilities, and access to private capital and government policymakers.²⁶⁰ The UK's excellent scientific and research base produces almost one-third of all European scientific journal publications and leads peers in Europe in total life sciences publications.²⁶¹ The nation also leads European peers in total new biotechnology start-ups and funding. Between 2018 and 2020, VC and IPO funding in the UK totaled around \$3.3 billion, dwarfing the next-highest nation—Switzerland.²⁶² The country also saw twice as many biotechnology companies founded, compared to peers in Switzerland and France.²⁶³

According to a 2021 analysis from McKinsey & Company, the UK beats the United States, the EU, and China in biotechnology-related scientific publications and approved patents.²⁶⁴ The same analysis places the UK second, behind only the United States, in overall biotechnology innovation.²⁶⁵

The biotechnology sector is a thriving and highly productive part of the UK economy. Life sciences is among the country's most successful sectors, valued at roughly \$132 billion in 2021.²⁶⁶ Two of the country's largest companies are biopharmaceutical companies—GlaxoSmithKline (GSK) and AstraZeneca, the latter achieving global acclaim for its breakthrough COVID-19 vaccine.²⁶⁷

UK biotechnology leadership is underpinned by strategic policies, investments, and initiatives. In 2011, the government introduced a *Strategy for UK Life Sciences*, which aspired to make the nation the “global hub for life sciences” with a focus on building a collaborative life sciences ecosystem, expanding the talent pipeline, and facilitating innovation in healthcare.²⁶⁸ Despite stiff competition in life sciences, the UK has fared well.

The UK is now considered a world leader in genomic research.²⁶⁹ This claim rests largely on the success of the UK Biobank, arguably the nation's most impressive national asset for biotechnology leadership. The UK Biobank, which began as a public-private partnership

in 2006, is a large-scale biomedical database that includes the fully sequenced genetic codes of 500,000 volunteers.²⁷⁰ The biobank makes this wealth of genetic data available to bona fide academic or commercial researchers to advance research in the public interest. The UK Biobank is a critical national asset for biotechnology R&D, especially in health applications.

In 2013, the UK also launched the 100,000 Genomes Project. By 2018, the project had fully sequenced 100,000 genomes from National Health Service patients, with the hope of creating a large enough genetic database to drive further research and innovation.²⁷¹ The project also resulted in the National Genomic Research Library, which includes de-identified samples, genomic data, and other health data that researchers can securely access.²⁷²

In recent years, the UK government has introduced several initiatives to reinforce its biotechnology leadership. In 2021, the government introduced its *Life Sciences Vision* outlining a national industrial strategy to secure UK leadership in what it called one of “the most valuable and strategically important” sectors for the economy.²⁷³ Among other steps, the *Life Sciences Vision* called for developing the UK’s science and clinical research infrastructure and better leveraging its genomic and health data.

The UK government has also funded the Catapult Network—a national network of cutting-edge R&D infrastructure from laboratories to testbeds to factories that helps bridge the gap between research and business-critical technologies, including cell and gene therapy and medicine discovery.²⁷⁴ The UK’s Advanced Research and Invention Agency, which supports high-risk, high-reward breakthrough technologies, has also invested in neural-machine interfaces and plant engineering.²⁷⁵

In 2023, the government published a *National Vision for Engineering Biology*, which laid out policy direction for government investment, regulation, and reform.²⁷⁶ The *National Vision* outlined several steps, including a roughly \$2.6 billion investment over the next decade, to drive engineering biology R&D.²⁷⁷ Later that year, the government announced a nearly \$840 million investment through the Life Sci for Growth package that unites 10 different policies, many of which will fortify the UK biotechnology sector.²⁷⁸ This includes nearly \$200 million to upgrade the UK Biobank and \$320 million to incentivize pensions to invest in promising technology firms.²⁷⁹ More broadly, the UK government has increased public spending on R&D for FY2024–2025 to its highest level ever, at roughly \$26 billion.²⁸⁰

In February 2024, UK Research and Innovation (UKRI), a nondepartmental public body supported by the government, also launched a roughly \$130 million fund to spur R&D in engineering biology.²⁸¹ The fund will support six new UKRI Engineering Biology Mission Hubs nationwide, which will receive up to \$15 million over five years.²⁸²

Although the UK is, in many ways, a demonstration of how renewed investment and strategic policy can bolster biotechnology competitiveness, its biotechnology sector is not without challenges. A relative lack of sophistication in biotechnologies in the UK’s VC community compared to the United States, for instance, constricts capital to promising biotechnology start-ups. Although the UK has longtime strengths in genetic research and in cell and gene therapy, it faces significant gaps in synthetic biology, hence the renewed push for government support. Bureaucratic approval processes for private or commercial research funding also hamper innovation. Nevertheless, the UK has not just talked about prioritizing biotechnology; its slew of policies, initiatives, and investments demonstrates a resolve to act.



Biopower: The United Kingdom

Tailwinds

- Harnessing biodata through the UK Biobank and 100,000 Genomes Project
- Detailed national strategies for life sciences and engineering biology
- Policy incentives and public support for biotechnology R&D, hubs, and private investment

Headwinds

- Slow approval processes for private and commercial research
- VCs less familiar with biotechnologies
- Relatively lagging in synthetic biology

Japan

Japan offers an interesting case of a government embracing the biorevolution as an opportunity to solve urgent social problems and reinvigorate its technology leadership. With the world's highest proportion of elderly persons and one of the highest longevity rates, the government faces a daunting rise in health care expenditures and questions about how to tap the strong domestic demand for new medical treatments and pharmaceuticals. (Japan is the third-largest pharmaceutical market in the world.²⁸³) At the same time, the government seeks to boost its relatively modest annual economic growth rates, fortify its critical health care and pharmaceutical sectors, and continue burnishing its global leadership in fighting climate change and promoting environmental sustainability. Tokyo views biotechnology as a potential solution to all these problems—lowering health care costs through new treatments, increasing economic growth through a rejuvenated biopharmaceutical industry, and protecting the environment through more sustainable biobased materials.

Japan enters the biorevolution with many strengths. It has the third-highest R&D spending on health among OECD nations and boasts major health care companies, such as Astellas Pharma, Eisai, and Takeda.²⁸⁴ The nation's pharmaceutical sector is nearly \$83 billion and growing.²⁸⁵

Japanese companies also enjoy a reputation for trust, quality, and reliability—all essential qualities for sensitive biotechnology research and applications. Japan has long pioneered key biotechnology-related applications, such as CAR T-cell therapies, which transformed cancer treatment.²⁸⁶ The nation is also a leader in regenerative medicine and gene therapy research. Japan boasts the largest bioinformatics database in Asia.²⁸⁷ Its biotechnology R&D focuses mostly on pharmaceutical and medical applications, although industrial and agricultural applications have earned renewed attention.²⁸⁸

Japan once led the world in biopharmaceuticals. In the 1980s, Japanese companies introduced 29 percent of all new chemical entities approved by the FDA.²⁸⁹ Between 2011 and 2020, that share fell to just 7 percent.²⁹⁰ The government is now working to reinvigorate the sector with several initiatives. In 2019, Japan laid out its Bioeconomy Strategy with ambitions to be “the world's most advanced bioeconomy” by 2030.²⁹¹ The strategy rests on three pillars: expanding the market for bio-products, fostering “bio-communities” across Japan (essentially, clusters of scientific, technological, and business talent), and promoting data linkage to enable cross-sectoral biotechnology innovation.²⁹²

In June 2022, the government of Japan also launched the Grand Design Action Plan for a New Form of Capitalism—the Kishida administration's industrial policy to both promote economic growth and solve social problems.²⁹³ The initiative named biomanufacturing, regenerative medicine, and gene therapies as key focus areas.²⁹⁴ The same year, the government announced \$700 million to increase support for start-ups by more than 10-fold.²⁹⁵ In 2023, the Japanese Ministry of Economy, Trade and Industry also announced \$3 billion over the next decade to promote the biotechnology ecosystem, with a focus on supporting Japanese start-ups in drug discovery, regenerative medicine, cellular therapies, and gene therapies.²⁹⁶

Japan's biotechnology sector also faces considerable challenges. In particular, the country's start-up scene remains underdeveloped and starved for capital. Last year, Japanese VC and corporate investment in biotechnology companies was barely 3 percent of comparable investments in the United States.²⁹⁷ Other challenges include a risk-averse business culture, complex regulatory regimes, slow approval processes, and biotechnology innovations concentrated in a few major companies. Japan may have considerable advantages, but it also faces significant headwinds to realize its potential.



Biopower: Japan

Tailwinds

- Proactive government strategy and significant public investment
- High public R&D spending, especially on health applications
- Robust health and pharmaceutical industry, with reputation for trust and quality

Headwinds

- Complex regulatory regime and slower approval processes
- Biotechnology innovation concentrated in a few large companies
- Risk-averse business and investment culture
- Start-ups struggle to access capital

South Korea

South Korea has emerged as a global leader in biotechnology, driven by its strong research and innovation, cutting-edge technology and medical infrastructure, highly educated workforce, and forward-thinking government policy. South Korea boasts a strong research and innovation base. Today, South Korea is the second highest among OECD countries for domestic spending on R&D.²⁹⁸ Broader government R&D investment in biotechnology specifically increased from \$1.2 billion in 2007 to \$3.4 billion in 2016. It has continued thereafter, with the total share of government R&D devoted to biotechnology rising from 15.7 percent to 19.2 percent from 2016 to 2020.²⁹⁹ The investments have paid off; today, South Korea ranks second globally for biomedicine production. Its biotechnology workforce continues to grow, along with its biotechnology exports, which doubled between 2016 and 2018.³⁰⁰

South Korea's robust biotechnology sector benefits from a vibrant technology innovation ecosystem of research institutes, universities, and major private companies at the cutting edge of technology innovation. Pillars of this ecosystem include Seoul National University's Institute of Molecular Biology and Genetics, which conducts cutting-edge life science research, and the Korea Advanced Institute of Science and Technology, which focuses on synthetic biology and biomedical engineering.³⁰¹ In addition, the Korea Research Institute of Bioscience and Biotechnology—a government-supported national think tank—conducts research in advanced biotechnologies and provides education and training programs, commercialization support for subject matter experts, and public infrastructure in the form of state-of-the-art research facilities.³⁰²

The innovation-friendly ecosystem has fueled South Korea's biotechnology sector. South Korea's share of patents filed under the Patent Cooperation Treaty in biotechnology industries grew from 4.2 percent of the OECD total in 2009 to 7.5 percent in 2017.³⁰³ The country also enjoys a robust start-up ecosystem, especially compared to its neighbor Japan. Between 2010 and 2016, the number of Korean biotechnology start-ups rose from 140 to 440—a more than three-fold increase.³⁰⁴ The Korean government has actively supported start-ups, spending \$1.2 billion in 2020 after accounting for total direct and indirect investment.³⁰⁵

In addition to a robust start-up scene, South Korea has several major biotechnology firms. Samsung BioLogics has emerged as a global leader in contract development and manufacturing organizations (CDMOs) and made

significant investments in cell-line development and clinical manufacturing.³⁰⁶ LG Chem has made significant investments to develop innovative drugs for complex diseases such as diabetes and cancer.³⁰⁷ SK Bioscience, which earned global attention for its work manufacturing COVID-19 vaccines during the pandemic, is also expanding its biomanufacturing capacity to produce next-generation vaccines at scale.³⁰⁸

In the last five years, the South Korean government has introduced a raft of biotechnology-related policies. In May 2019, it introduced the Innovative Strategy for the Biohealth Industry, which outlined holistic support for the biohealth sector from R&D to commercialization.³⁰⁹ Specifically, it called for collecting biodata from up to one million Korean volunteers, which would be stored in a new National Biological Resources Center.³¹⁰ The strategy also called for tax exemptions, loan support, shorter review periods, and a greater than 50 percent increase in total R&D investment in biotechnology between 2017 and 2025.³¹¹

In November 2022, the Ministry of Science and ICT launched the National Synthetic Biology Initiative to boost synthetic biology and biomanufacturing through several high-level ambitions, such as transitioning 30 percent of the country's manufacturing industry to biomanufacturing within a decade.³¹² To do this, it committed government assistance to train 1,000 synthetic biology professionals, establish multidisciplinary departments that combine biology with AI and engineering, and create a graduate school focused on synthetic biology, among other measures.³¹³

In April 2024, the Korean Presidential Advisory Council on Science and Technology laid out a plan to make South Korea “an advanced biotechnology powerhouse” by 2035 through a focus on “digital-bio convergence” and biomanufacturing innovation.³¹⁴ The comprehensive plan has many facets, including ambitions to make South Korea a leader in digital biotechnology—or the intersection of data, AI, and biotechnology—and shifting the nation's petrochemical-based materials industry toward a biobased industry through a national network of biofoundries.

Despite significant strengths, South Korea faces several challenges. Like Japan, South Korea must do more to promote a culture of entrepreneurial risk. According to a 2019 survey by the World Economic Forum on “Attitudes Toward Entrepreneurial Risk,” South Korea ranked 88 out of 144.³¹⁵ Another survey of 19 high-income countries ranked South Korea 17th for a perceived lack of opportunities for starting a business.

Although the government offers generous assistance for biotechnology start-ups, excessive paperwork can

discourage firms from accessing it.³¹⁶ Total start-up financing reached \$31 billion in 2021 but remains short of total demand.³¹⁷ Combined with regulatory hurdles, Korean start-ups face difficult exit models: only 25 percent of all VC investment in Korean start-ups is recouped through IPOs or M&As, compared to 94 percent in the United States.³¹⁸ Similarly, the Korean government provides generous funding for late-stage technology R&D and commercialization, but its support for early-stage research remains insufficient. Just 20 percent of government R&D funding goes to basic research, according to a study from Korea Institute of Science and Technology Evaluation and Planning.³¹⁹

The Korean government has made a longtime bet on its biotechnology leadership, with a long track record of plans, investments, and initiatives to promote it. The country has also made a strategic decision to carve out a niche for itself on the world stage at the intersection of biotechnology, health, AI, and digital technologies. Time will tell if it pays off.



Biopower: South Korea

Tailwinds

- Strong research and innovation base with high domestic spending on R&D
- Growing public investment in biotechnology R&D
- Growing biomedical production, workforce, and exports
- Cutting-edge technology and medical infrastructure with world-leading firms such as Samsung BioLogics
- Forward-thinking government policies to leverage biodata, improve financial support, reform regulations, increase public investment, and position the country as a specialist in “digital-bio convergence”

Headwinds

- Risk-averse culture that can discourage entrepreneurship
- Inadequate financing for biotechnology start-ups with difficult exit models
- Insufficient support for early-stage research

China: The Rising Biopower

China is racing to bypass the United States as the global biopower, and it has the infrastructure, investment, policies, and political will to do so. Beijing has shown an interest in biotechnology as early as 1986, when Chairman Deng Xiaoping first created the 863 Program to reduce China’s foreign dependence on strategic technologies and to bolster its indigenous technology capabilities, including for biotechnology applications in medicine and agriculture.³²⁰ The 863 Program elevated biotechnology leadership as a national priority, increased R&D funding, and began laying a foundation of biotechnology R&D infrastructure. In recent years, biotechnology has gained more prominence in China’s broader push to dominate key emerging technologies.

China’s biotechnology sector has enjoyed several natural advantages, including access to relatively cheap labor; a large, growing, and increasingly skilled talent pool; and significant expected demand from a massive domestic market.

As with several other emerging technologies, China has nearly closed the biotech gap with the United States since the 1980s through both licit and illicit means. In an infamous 2013 case, U.S. authorities indicted six Chinese nationals for conspiring to steal GM corn varieties.³²¹ According to a report prepared for the U.S.-China Economic Security Review Commission in 2019, Beijing has benefited from the relatively open and collaborative period in U.S.-China relations in the two decades following its accession to the World Trade Organization, which enabled joint R&D projects, trade, and talent exchanges.³²² Over that period, Chinese firms acquired key technologies through outright purchases, M&A, partnerships, and forced transfers as a condition of market entry for foreign firms. In addition, China’s biotechnology sector has enjoyed several natural advantages, including access to relatively cheap labor; a large, growing, and increasingly skilled talent pool; and significant expected demand from a massive domestic market.

Beyond these advantages, the country’s biotechnology sector has benefited from ambitious long-term policies at the central, regional, and even municipal levels. Beijing

has embraced a comprehensive, long-term strategy to foster, protect, and mature the country's biotechnology sector. It has done this through several overlapping, self-reinforcing lines of effort: boosting basic and applied R&D, developing world-class research infrastructure, expanding domestic biomanufacturing capabilities, and cultivating talent.³²³

China has made considerable strides in becoming a global leader in biotechnology R&D. In 2021, an index from *Nature* magazine found that China hosts 8 of the world's 100 best life science institutes.³²⁴ However, it lags significantly behind the United States, which has 51 such institutes, and Europe, which has 28.³²⁵ Nevertheless, according to the ASPI Critical Technology Tracker, which measures a country's technology leadership by its share of "high-impact research," China now leads the world in several key biotechnologies—specifically biofuels, synthetic biology, and biological manufacturing.³²⁶ It ranks second only to the United States for vaccines and medical countermeasures.³²⁷ China's dominance is especially pronounced for synthetic biology, in which its share of high-impact research is more than triple that of the United States, posing a high risk of monopolization.³²⁸ Similarly, China's share of high-impact research for biofuels is more than four times the share of the United States.³²⁹ Chinese-affiliated scholars also rank second globally for authoring biomedical papers, albeit with a lower share than leading competitors in top journals.³³⁰

China also enjoys a strong and growing biotechnology innovation ecosystem and manufacturing capabilities. As in most countries, China's biotechnology sector has overwhelmingly focused on medical and health applications given their large expected demand and value for the country's aging population.³³¹ Favorable policies and regulatory reforms from Beijing, a large talent pool, and significant capital investments have also fed rapid growth of the country's biomedical sector in particular.³³² In recent years, China has worked hard to align with global regulatory standards, most notably testing and approval requirements for the FDA and European Medicines Agency (EMA), allowing it to expand beyond its sizable domestic market into lucrative developed economy markets abroad.³³³

As in other sectors, China's low-cost, high-quality manufacturing has allowed it to dominate key segments of the biomedical sector. Today, China is the largest exporter of active pharmaceutical ingredients to the U.S. market, and they routinely earn FDA and EMA approvals.³³⁴ As an example, China exports roughly 40 percent of the world's active pharmaceutical ingredients.³³⁵ The

country's top CDMOs and biotechnology firms also account for roughly 7 percent of global capacity for monoclonal antibodies.³³⁶

Still, China's biomedical and pharmaceutical sector remains relatively low on the value chain and focused principally on biologics. Contract research and manufacturing organizations still represent a significant share of China's biotechnology sector, and Chinese biomedical firms have become global sources for biosimilars—essentially, cheaper reproductions of original biological drugs—even as they lag in breakthrough innovations. However, Beijing has ambitions to move up the biologics value chain and deliver cutting-edge breakthroughs in areas such as gene editing through cutting-edge CAR T and CRISPR techniques. To that end, Chinese biopharmaceutical firms have accelerated innovation in recent years.³³⁷ The government has also designated the biopharmaceutical industry as strategically important, creating a power signal for markets and political support to mature the sector's capabilities.³³⁸

Until recently, relatively unrestricted investment flows have helped China close the biotechnology gap with the United States. Inbound investment, largely in the form of foreign direct investment (FDI) and VC funding, helped nourish Chinese biotechnology firms, integrate them into global supply chains, and transfer valuable expertise and intellectual property.³³⁹ As Chinese biotechnology firms matured, they began investing externally through acquisitions, greenfield investments, and VC plays of their own. According to a 2019 report for the U.S.-China Economic and Security Review Commission, Chinese outbound investments in the pharmaceutical and biotechnology sector reached \$1.5 billion in 2015 and more than doubled to \$3.5 billion in 2017.³⁴⁰ Chinese VC investment in the global biotechnology industry has risen steadily over the past decade as well, concentrating in North America, Europe, Singapore, and Australia.

At the same time, FDI continued to pour into Chinese biotechnology firms. According to an analysis by BCIQ, between 2018 and 2020, VCs invested \$12 billion in Chinese biotechnology firms, triple the amount between 2015 and 2017.³⁴¹ By early 2022, only the United States had greater market capitalization than China for biotechnology companies launched since 2015.³⁴² China has also benefited from overseas training of Chinese students and researchers, who then repatriate to China with strong incentives from initiatives such as the Thousand Talents Program.³⁴³ According to the Center for China and Globalization, a Beijing-based think tank, there were more than one million Chinese postsecondary students abroad in 2021, making China the world's largest source

of international students.³⁴⁴ Many Chinese nationals studying in the United States are returning home—in part due to increased opportunities in China, but also because of growing U.S.-China tensions and the perception of a shrinking “ceiling” for career advancement. Indeed, departures of U.S.-based Chinese scientists have increased by 75 percent since 2018, with two-thirds relocating to China.³⁴⁵

China has also actively supported its domestic biotalent. As one data point, Chinese schools have dominated recent iGEM competitions. In 2023, Chinese high schools won 42 gold medals, nearly three-fourths of all gold medals awarded.³⁴⁶ U.S. high schools won three gold medals. Chinese schools won 45 gold medals at the undergraduate level, compared to 10 for the United States. Chinese iGEM teams, like any team in the competition, receive support from a range of sponsors, including government-backed national laboratories, foundations, and R&D programs.³⁴⁷

Beijing’s Biotechnology Policies

Beijing has long championed the country’s biotechnology sector through ambitious, long-term policies. Biotechnology first featured in China’s seventh Five-Year Plan, covering the period from 1986 to 1990, and has gained prominence ever since.³⁴⁸ Beijing named biotechnology a “strategic emerging industry” in 2013 and again in 2020.³⁴⁹ China’s 13th Five-Year Plan upped the ante, calling for transitioning China from a biotechnology power to a biotechnology superpower.³⁵⁰

In an April 2022 speech, Chinese Communist Party General Secretary Xi Jinping urged Chinese scientists to overcome “bottlenecks” and deliver breakthroughs in life sciences, and China’s latest State Council work report emphasized biomanufacturing and life sciences as key industries under China’s “new productive forces” initiative to encourage new engines of economic growth and development.³⁵¹

In the latest Five-Year Plan, covering 2021–2025, Beijing added even more policy heft with several specific ambitions for its biotechnology leadership:³⁵²

- Establish a national laboratory focused on biotechnology and pharmaceuticals
- Establish a national biological data center system and integration of biotechnology and information technology more broadly
- Accelerate development of bioenergy
- Develop stronger biosafety governance and risk monitoring
- Establish an R&D risk assessment mechanism for biotechnology

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The latest Five-Year Plan also calls for “strategic scientific plans and scientific projects” in seven fields, three of which relate to biotechnology: brain science and brain-inspired research, genetics, and clinical medicine and health.³⁵³ Biotechnology is also prominently featured in the Made in China 2025 (MIC 2025) plan, which aims to make the country a global leader in advanced manufacturing.³⁵⁴

Beyond the five-year plans, China has simultaneously introduced a raft of specific, biotechnology-related national strategies, including to drive medical AI development and integrate biotechnology and AI.³⁵⁵ China has boosted its biomedical sector with policies for clinical trials, procurement rules to ensure a large domestic market, generous R&D subsidies, export financing, high-level diplomatic and political backing, and process reforms such as fast-track approvals for Chinese-developed drugs.³⁵⁶ Preferential tax policy for high-tech start-ups, R&D expenses, and accelerated depreciation for fixed assets also provide an advantage.³⁵⁷ Similar regional and municipal policies complement these advantages, while offering additional forms of support such as subsidized certification costs, land guarantees, and accelerated customs clearings.³⁵⁸

Another driver of China’s biomedical sector is a national network of biopharmaceutical industrial parks, supported by both central and provincial governments.³⁵⁹ These parks foster, accelerate, and mature domestic biopharmaceutical capacity and development. Industrial parks have emerged as one of Beijing’s favored means of developing strategic sectors; by the end of 2019, China boasted 387 such industrial zones, 193 of which focused on the pharmaceutical sector—50 percent.³⁶⁰

In recent years, China has also begun to shift from its longtime skepticism of genetically engineered (GE) and GM agricultural products, likely due to a growing interest in ensuring food security amid increased geopolitical tensions.³⁶¹ Historically, China has not produced significant GM crops—except for cotton—due to onerous regulations and limited domestic demand. Except for cotton and papaya, Beijing has yet to approve any GE food or feed products for domestic cultivation.³⁶²

However, Beijing has since reformed its regulations to lessen regulatory burdens and promote commercial cultivation of GE crops.³⁶³ It issued its first-ever regulations for GE plants and is developing rules for GE products more broadly, while increasing investment in GM-related research.³⁶⁴

Following a similar playbook to other strategic sectors, such as telecommunications, Beijing has also employed concerted policies to promote and protect select national biotechnology champions, such as BGI Group and WuXi Biologics, creating an unfair playing field that allows them to dominate the vast domestic market and mature into serious global players.³⁶⁵ A recent report from the Center for Security and Emerging Technology details how Beijing has applied familiar industrial policy to dislodge Western leadership in gene sequencing—a foundational capability for a range of biotechnology breakthroughs.³⁶⁶ Through many of the policies described here, Beijing has cultivated BGI as the “pointy end of the spear” for its biotechnology ambitions, with the goal of dominating not only gene sequencing equipment but also gene sequencing services.³⁶⁷ In 2010, a state development bank gave BGI \$1.5 billion to support its R&D.³⁶⁸

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Tailwinds

Arguably, China’s biggest biotechnology advantage over the United States is biodata. Indeed, China is poised to become the world’s biodata leader—a position that could confer significant advantages for its already fast-maturing biomedical and biopharmaceutical sector.³⁶⁹ Early on, Beijing understood the strategic value of biodata and has proactively amassed it through domestic policy and global harvesting using both legal and illegal means.

China sees the value of large-scale biodata in unlocking breakthroughs for health and medicine. Accordingly, China’s State Council views big data in health and medicine as a national priority and has moved to build national and regional data centers to store large-scale health data in areas such as Fuzhou and Changzhou. These data centers will also complement the government’s \$9.2 billion effort to boost precision medicine.³⁷⁰

Several domestic policies feed China’s rapid accumulation of biological and genetic data. For one, the country’s current Five-Year Plan for National Informatization requires hospitals to use digital health records.³⁷¹ Another initiative, Healthy China 2030, calls for standardization of biodata across data centers and promotes making such large-scale data more accessible to researchers to unlock new biomedical applications.³⁷² The country’s Biosecurity Law makes it clear that Chinese biodata falls within the government’s sovereignty, while another policy calls for the consolidation of national security-relevant scientific data produced within China into new data centers.³⁷³

China has also endured global criticism for its mass collection of biodata—either forcibly, or without informed consent—from residents in highly oppressed regions such as Xinjiang and Tibet, allowing it to diversify a biodataset otherwise dominated by Han Chinese.³⁷⁴ BGI has also come under criticism for collecting biodata abroad through its noninvasive prenatal tests, which it sells around the world, even as reports have surfaced that the company has collaborated in the past with the People’s Liberation Army and shared data from such tests with China’s National GeneBank (CNCB), which BGI runs.³⁷⁵ BGI has denied these allegations. Today, the CNCB houses more than 10 million genetic data samples.³⁷⁶

Another criticism toward China is that, while its researchers can access the world’s biodata through open-source datasets such as the UK Biobank and NIH’s NCBI, this access is not reciprocal. Chinese national security and data privacy laws directly restrict access to Chinese genetic data without government approval.³⁷⁷ In other

words, this policy imbalance benefits Chinese biotechnology researchers by allowing them access to far more biodata than their U.S. counterparts. Left unaddressed, this imbalance in data access could lead to gaps in biotechnology capabilities down the road.³⁷⁸

Headwinds

Although China has considerable advantages in its bid for biopower, it too faces headwinds. Growing geopolitical uncertainty could mean further restrictions on Chinese biotechnology products—including its fast-growing biologics sector—in developed markets, stunting its growth potential and revenue to reinvest in R&D and stay globally competitive. Continued progress by Chinese biomedical firms to integrate with global regulatory and approval processes is necessary for sustained growth but would also likely become imperiled should geopolitical tensions intensify. Although China's biotechnology talent pool is growing and has made strides in attracting foreign talent, it still struggles to compete with top destinations for STEM talent—notably, the United States.³⁷⁹

Growing geopolitical tensions could also limit the two-way capital flows that have helped mature China's biotechnology sector, by driving down FDI and VC funding into China through increased risks and outright restrictions, and by limiting the ability of Chinese firms to make outbound investments in developed, biotechnology-leading markets such as the United States, Europe, and Japan. This challenge will only grow as the United States implements the August 2023 executive order tightening screens on outbound investment into national security–relevant technologies.³⁸⁰ In a recent indication of trouble, data from ITjuzi found that financing for pharma and biotechnology start-ups dropped by 60 percent between 2023 and 2021.³⁸¹ Even more ominous, the number of Chinese start-ups has collapsed from a high of 51,302 in 2018 to just 1,202 in 2023.³⁸²

China's biopharmaceutical sector may continue to lag in clinical development due to misalignment with global standards and regulations. For instance, to approve new treatments, Western regulators typically require a higher standard of data disclosure, security, and care than many Chinese firms can meet—although they are closing the gap. Western regulators often require data from patient populations that demographically mirror their own, which limits the value of relatively homogeneous China-only clinical trials. If this trend continues, it will inhibit innovation and impede China's ambitions to move up the biomedical value chain, continuing its trend of focusing on late-stage development instead of the higher-risk, higher-reward early-stage development. Even if China

were to boost its biomedical innovation, it still faces challenges in building a mature, global ecosystem to commercialize and scale breakthroughs, owing to challenges with capital, talent, and access to global markets.³⁸³

Forecasting China's Biotechnology Leadership

Despite headwinds, China's biotechnology sector has achieved massive growth in recent years, even if it remains just a fraction the size of the United States' in terms of market capitalization.³⁸⁴ In 2019, for instance, China's biotechnology industry was just one-tenth of the United States' sector.³⁸⁵ Having matured, China's biomedical and biopharmaceutical sectors are now looking beyond China to global markets and are poised to play a significant role in the global biopharmaceutical market.³⁸⁶ Although China's biopharmaceutical sector still has a considerable gap to close in innovative R&D, it is making steady progress. In the past five years, total innovative assets under clinical development in China have tripled.³⁸⁷ At the same time, China is contributing a growing share to global biomedical research, even as it lags in top-tier publications and breakthrough research.³⁸⁸ A McKinsey & Company analysis predicts that Chinese biopharmaceutical companies will continue to lead in “innovations in oncology, monoclonal antibodies, and small molecules” through 2028.³⁸⁹ If Chinese biomedical firms can maintain price competitiveness as they expand into new modalities, they could secure valuable market share in emerging economies from Latin America to Southeast Asia.³⁹⁰

Having matured, China's biomedical and biopharmaceutical sectors are now looking beyond China to global markets and are poised to play a significant role in the global biopharmaceutical market.

At the same time, China will likely continue to hold its position as a global biomedical supply hub, owing in part to its impressive domestic manufacturing capabilities for biopharmaceutical products. Beijing's success at amassing biological and genetic data—and the data imbalance it has erected for foreign competitors—will remain another powerful advantage.³⁹¹

In sum, no country presents a greater challenge to U.S. biotechnology leadership than China. The outcome will hinge on whether the sheer brute force of China's

policies, infrastructure buildout, talent development, and public investment will overcome longtime shortcomings in cutting-edge research, innovation, and commercialization, as well as new challenges from rising geopolitical tensions, falling biotechnology investment, and a faltering ecosystem for technology start-ups. Just as important, the outcome will depend on whether U.S. policymakers coast on the nation's current leadership or take action to secure it for a generation.



The Rising Biopower: China

Tailwinds

- Several long-term biotechnology-related national strategies, including a strategy to integrate biotechnology and AI
- Preferential tax, land, and other policies for biotechnology, along with generous R&D subsidies, export financing, high-level diplomatic and political backing, and fast-track approvals for Chinese-developed drugs
- Large-scale, accessible biodata amassed through licit and illicit means
- Rapidly growing biomedical and biopharmaceutical sectors
- Expanding share of global biomedical research
- Status as a global biomedical supply hub due to robust domestic manufacturing
- National network of biopharmaceutical parks

Headwinds

- Biopharmaceutical sector that trails in innovative R&D
- Lags in top-tier biomedical publications and breakthrough research
- Rising geopolitical tensions that could impede inbound capital, slow the maturation of the biotechnology ecosystem, and stifle expansion in developed markets
- Collapsing biotechnology and pharma investment and rate of technology start-ups

Chapter 05

Recommendations

The question in the coming years is whether policymakers and industry play America's hand wisely or surrender its leadership to China and other foreign competitors, with potentially grave costs to U.S. national security and economic interests. Although the Biden administration has made notable progress in elevating biotechnology as a strategic priority and begun construction of a policy framework with its biotechnology EO, significant work remains. Securing America's position as the global biopower requires that policymakers act now to accelerate the country's natural headwinds, including an unrivaled capital market to seed and scale biotechnology firms and applications; an unmatched ability to generate and attract top biotalent; and an unprecedented global network of partners and allies bound by shared democratic values. At the same time, policymakers must navigate growing headwinds that could stall progress. These include a still-immature policy framework; nonexistent or needlessly complex federal definitions, standards, and regulations; inadequate, imbalanced, and poorly coordinated public investment, combined with private investment that overwhelmingly focuses on biopharmaceutical applications

at the expense of other sectors; uncertainty over how best to unleash the nation's biodata while ensuring respect for privacy and civil liberties; and enduring public skepticism over biotechnology applications in ethically fraught areas such as gene editing. This section outlines several immediate and longer-term recommendations for leaders in policymaking and industry in six key areas:

- Invest in research and infrastructure
- Expand biotalent
- Unlock biodata
- Strengthen and streamline policy
- Lead globally
- Build public trust

The recommendations below would require roughly \$20 billion in new federal investment to help secure American biopower for a generation. Policymakers should view that funding level as the floor. A more detailed breakdown of this \$20 billion investment can be found in the appendix.



Invest in Research and Infrastructure

No country matches America's combined public and private investment in biotechnology. However, public investments diffuse across multiple agencies and programs, limiting their impact and accessibility. Public investment in life sciences R&D flows overwhelmingly through NSF and NIH, mostly supporting health and medical sectors and dwarfing initiatives to support the broader bioeconomy such as BioMADE, T-BRSC, DBMIP, and ARS. Taken together, federal investments fall short of the levels required to close gaps with China in key fields, specifically synthetic biology and biological manufacturing, genomes and genetic sequencing and analysis, and novel antibiotics and antivirals.³⁹² In addition, public investments remain inadequate to build a robust domestic bioindustrial base, which is virtually nonexistent today.

The United States leads the world in private investment in biotechnology, but investments remain overwhelmingly concentrated in the biomedical and biopharmaceutical sectors. This leaves many ventures in broader biotechnology applications for defense, agriculture, energy, and manufacturing starved of capital and infrastructure. This dynamic has left other promising biotechnology applications undercapitalized, underdeveloped, and vulnerable as firms cross the Valley of Death—the perilous time when their initial funding runs low, and they require more capital to scale and survive. Many private investors recognize the potential of broader biotechnology applications but view their returns as more distant and uncertain. Policy can incentivize private capital in broader biotechnology applications through direct public investment, increased demand through stronger biobased procurement policy, and other incentives such as preferential tax treatment and loan guarantees.

Specifically, policymakers should adopt the following recommendations.

Expand infrastructure to seed, support, and scale biotechnology ventures across the Valley of Death

Congress should create incentives to attract private investments to national security–relevant biotechnologies. For instance, Congress should strengthen the R&D tax credit for key national security–relevant technologies, including biotechnology, and consider deductions for accelerated depreciation of R&D expenses. In both cases, Congress could limit these tax benefits to early-stage biotechnology firms to limit costs. Congress should protect and ideally extend the Qualified Advanced Energy Project Credit, which provides a 30 percent tax credit for many bioenergy and bioindustrial projects.³⁹³ Finally, Congress should examine the UK's Long-Term Investment for Technology and Science initiative, which has provided up to \$320 million in government support to create mechanisms for large institutional investors, such as universities and pension funds, to invest in science and technology firms.³⁹⁴

Boost demand for biobased products by further leveraging federal purchasing power

The U.S. government should flex its purchasing power to increase long-term demand for biobased products, spurring innovation and fortifying the biotechnology ecosystem during economic downturns—especially in sectors beyond health and medicine. Today, the USDA leads the BioPreferred Program, which requires that the federal government and its contractors give purchasing preference to biobased products for certain categories of products, with some exemptions.³⁹⁵ Currently, the USDA has designated 139 categories, ranging from paints to household cleaners, under the program.³⁹⁶ For all undesignated categories, the federal government must give purchasing preference to products that contain at least 25 percent biobased content, with some reasonable exceptions. To boost demand for biobased products, the USDA should expand the list of designated products for the BioPreferred Program beyond the current total of 139. For all undesignated products, they should explore the feasibility of raising the minimum threshold from 25 percent to 50 percent over time. It should also partner with the General Services Administration to expand training for federal procurement officers to better understand the BioPreferred requirements and explore reforms to facilitate BioPreferred-compliant purchases. In addition, the new administration should direct the DoD to develop a plan to leverage its procurement power to boost

demand for biomanufactured products and materials and explore synergies with investments from T-BRSC, DBMIP, and the Office of Strategic Capital (OSC).

Direct BioMADE to allocate existing funds before it receives more

BioMADE is an independent, nonprofit membership body that seeks to foster a “sustainable, domestic end-to-end bioindustrial manufacturing ecosystem.”³⁹⁷ As of April 2024, BioMADE had only announced \$165 million in funded projects—just 36 percent of its \$450 million budget from the DoD.³⁹⁸ Conceptually, BioMADE holds great promise. Before Congress scales funding to the organization, however, BioMADE must prove that it can effectively allocate existing resources. The next administration should also clarify eligibility for project funding, including greenfield projects. According to the authors’ interviews with U.S. government personnel, ambiguity over project eligibility has delayed new funding awards.

Make biotechnology projects eligible for FAST-41 expedited federal environmental review and permitting

To expedite construction of biofoundries and other biotechnology-related infrastructure, Congress should make federally supported biotechnology projects eligible for the FAST-41 program, which currently considers expedited federal environmental review and permitting for 18 sectors, including AI, data storage and data management, cybersecurity, and quantum information sciences and technology—but not biotechnology. Given the relatively lower cost of biofoundries, Congress should consider reforms to include projects with total investment less than \$200 million, the current threshold for FAST-41 eligibility.³⁹⁹

Unleash the DOE’s data, computing, and research infrastructure for biotechnology

The DOE played a leading role in the Human Genome Project alongside NIH and boasts formidable, if largely untapped, resources for biotechnology. The DOE is the top generator of classified and unclassified scientific data. It boasts the world’s largest collection of advanced experimental facilities, including specialized facilities for genomics. It hosts the Advanced Biofuels and Bioproducts Process Development Unit at the Lawrence Berkeley National Laboratory.⁴⁰⁰ It has the world’s “fastest, most powerful, and highly energy efficient”

supercomputers, with open access to the scientific community. Together with its national labs, the DOE has a network of more than 40,000 scientists, including “chemists, biologists, materials scientists, and computer scientists.”⁴⁰¹ In July 2024, the DOE announced a major new initiative, the Frontiers in Artificial Intelligence Science, Security and Technology (FASST) initiative, which aims to “build the world’s most powerful integrated scientific AI systems” with a focus on data, computing infrastructure, applications, and model security and trustworthiness.⁴⁰² The next administration should direct the DOE’s Office of Critical and Emerging Technologies to develop a department-wide strategy for leveraging its diverse resources for biotechnology, from expanding support for workforce development, testbeds, and commercial partnerships, to opening computing and user facilities to vetted biotechnology researchers and ventures, to making the AI-biotechnology nexus a priority of the FASST initiative. The strategy should also identify funding and policy gaps.

Build a national biofoundry network

A robust network of regional biofoundries would allow researchers and industry to access advanced tools for bioengineering and biomanufacturing, driving cross-sector collaboration and innovation. To build this network, policymakers should significantly expand NSF’s BioFoundries program, which supports the construction of integrated facilities that combine biological disciplines with automation, AI, and advanced computing to drive biotechnology innovation.⁴⁰³ In August 2024, NSF announced awards totaling \$75 million to help establish five biofoundries across the country. Congress should double NSF’s available funding for this effort, allowing NSF to make at least \$150 million in awards annually to seed 100 biofoundries nationwide over the next decade.

Scale the DoD’s efforts to boost the defense bioindustrial base to \$6.35 billion

As part of the Biotechnology EO, the DoD committed \$1 billion for the DBIMP and \$270 million for T-BRSC over a five-year period from FY2022 to FY2026. T-BRSC is run by a program element in the U.S. Army and helps translate defense-relevant biotechnology research into products and to shore up defense supply chains by supporting the development of biobased materials. DBIMP has an overlapping but broader mandate and focuses on scaling defense-relevant biotechnology manufacturing capabilities after they have demonstrated value. The

next administration should better focus, coordinate, and elevate these efforts to ensure continued support for defense-relevant biotechnology products from research to prototyping to development to production, leveraging DARPA's Biological Technologies Office and the DoD's new OSC, which combines public and private investment to fortify defense-critical technology supply chains. Biotechnology is one of the OSC's 14 critical technology areas.⁴⁰⁴ Congress should double the DoD's total current spending of \$1.27 billion over five years, provided T-BRSC and DBIMP spend existing funds strategically. As of September 2024, DBIMP had announced 25 awards totaling \$42 million.⁴⁰⁵

Increase funding for DARPA's Biological Technologies Office

The administration's FY2025 budget request for DARPA is \$4.37 billion, an increase of \$247 million from the FY2024 enacted level.⁴⁰⁶ Congress should specifically increase funding for the BTO to support truly "high-risk/high-reward" breakthrough biotechnologies that support DoD operations and boost warfighter readiness.

Scale the Advanced Research Projects Agency for Health's funding to at least \$3 billion for FY2025, double the previously enacted level of \$1.5 billion

The Advanced Research Projects Agency for Health (ARPA-H), is a new independent body housed within NIH focused on high-potential, high-impact biomedical and health research.⁴⁰⁷ Most ARPA-H lines of effort involve biotechnology, including programs to drive breakthroughs in printing human organs, self-healing joints, engineering immune cells in the human body, precision cancer therapy, and early cancer detection.⁴⁰⁸ Increased funding would supercharge these efforts with significant potential benefits to U.S. leadership in biomedical innovation.

Allocate \$50 million annually through America's Seed Fund for biotechnology

NSF leads the Seed Fund, which awards more than \$200 million a year in R&D funding to roughly 400 technology-focused start-ups nationwide with "founder-friendly terms" but is routinely oversubscribed.⁴⁰⁹ Under the NSF program, technology-focused start-ups with the potential for significant positive social impact are awarded up to \$2 million—including up to \$305,000 in nondilutive

funding for R&D and prototyping.⁴¹⁰ The Seed Fund offers support for up to 18 months to help firms develop their prototypes and proof-of-concepts, followed by up to two years of support to enable technology transfer and commercialization.⁴¹¹

To catalyze America's biotechnology start-ups, Congress should authorize at least \$50 million in new resources for the Seed Fund to support biotechnology ventures for the next decade. Whereas broader NSF funding tends to support general biotechnology-related R&D at universities, America's Seed Fund ventures can help derisk private capital by signaling confidence in the venture and demand for its products.

Fund NSF's FY2025 budget request of \$10.18 billion and meet its biotechnology budget request of \$421 million⁴¹²

Currently, NSF invests around \$250 million annually to support U.S. biotechnology and the bioeconomy.⁴¹³ This funding supports biotechnology-relevant research grants in areas ranging from the molecular foundations of biology to plant genomes, synthetic cell design, and biomechanical engineering.⁴¹⁴ These grants largely flow to universities to support general and applied research related to biotechnology. The 2022 CHIPS and Science Act doubled NSF's total authorized funding to \$81 billion over five years from 2023 to 2027.⁴¹⁵ However, Congress has yet to obligate the total amount, leaving NSF's budget short \$8.5 billion in FY2023 and \$2.4 billion in FY2024.⁴¹⁶ In its proposed FY2025 budget, NSF requested \$10.18 billion total, including \$421.18 million for biotechnology.⁴¹⁷ Congress should view the \$421.18 million as the floor, not the ceiling, and ensure sustained funding at this level or higher in the years to come. This funding would accelerate NSF-supported biotechnology general and applied research, workforce training, and partnerships with the private sector to drive technology transfer and commercialization. It should also continue and expand the \$28 million for NSF's Future Manufacturing initiative, which provides grants of up to \$3 million in seed and research funds for biomanufacturing.⁴¹⁸ At the same time, Congress should direct NSF to develop a plan within a year to better consolidate and coordinate biotechnology investments across its directorates.

Fund biotechnology moonshots and *Bold Goals*

The 2023 *Bold Goals* identified five priorities for an ambitious, 20-year biotechnology research agenda.⁴¹⁹ These spanned from the DOE's goal to demonstrate and

deploy biobased alternatives that could replace more than 90 percent of existing plastics and other commercial polymers to the USDA's goal of reducing agricultural methane emissions to help reduce U.S. greenhouse gas emissions by 50 percent.⁴²⁰

The *Bold Goals* seek to galvanize public, private, and academic investment and collaboration, but they did not include any new funding or agency commitments to prioritize existing funds. The five lead agencies should identify the funding required to progress rapidly toward their *Bold Goals*. At a minimum, Congress can meet the DOE's budget request for \$36.54 million in new funding for biological and environmental research, along with a \$95 million increase for manufacturing and energy supply chains. To boost federal biotechnology research across government to support the *Bold Goals*, Congress should fund the \$50 million appropriated to the Agriculture Advanced Research and Development Authority (AgARDA) and increase funding for the Biomedical Advanced Research and Development Authority (BARDA) by \$20 million over FY2023 levels.⁴²¹ At the same time, Congress could task and fund agencies to pursue additional biotechnology moonshots, from artificial photosynthesis and fuel cells to universal vaccines to biobased carbon capture.

Fund NIH's FY2025 budget request and specify biotechnology priorities

In the most recent fiscal year, Congress reduced NIH's budget by \$368 million, the agency's first such reduction in a decade.⁴²² In its FY2025 budget, NIH requested \$50.2 billion, a \$2.4 billion increase over the prior fiscal year.⁴²³ Congress should match this request and restore FY2023 funding levels to the NIH Innovation Account to back several biotechnology-relevant goals from precision medicine to neurotechnologies from its FY2023 level of \$1.085 billion.⁴²⁴ In addition, Congress should also push NIH to provide funds beyond disease-specific research to include platform technology development, while specifying funds to renovate and build nonfederal biotechnology research facilities and boost relevant centers such as the National Institute of Biomedical Imaging and Bioengineering, whose funding has remained virtually flat for years.

Fund the DOE's FY2025 budget request for biotechnology-related priorities. The DOE has requested \$5 million in additional funding for its new Office of Critical and Emerging Technologies, which is charged with coordinating diffuse biotechnology investments across the DOE, among other roles. The DOE has also

requested level funding of \$280 million for its Bioenergy Technologies Office, which supports research to convert domestic biomass into biofuels and bioproducts.⁴²⁵

Fund NIST's FY2025 budget request to accelerate development of biotechnology standards and measures

NIST plays a leading role in development biotechnology measurements and standards, which are foundational to biotechnology research and policy. However, years of inadequate funding has left it with dilapidated facilities and antiquated information technology (IT) infrastructure. A 2023 NASEM report found that NIST's outdated infrastructure seriously impeded its ability to execute on its core mission, including an "inability to perform critical measurements in biotechnology."⁴²⁶ At a minimum, Congress should meet NIST's request for an additional \$311.5 million for FY2025 to modernize its infrastructure and support biotechnology-relevant initiatives such as the National Institute for Innovation in Manufacturing and Biopharmaceuticals (NIIMBL).



Expand American Biotalent

American biopower hinges on access to world-class biotalent in industry, academia, and government. As with most critical technologies, the United States faces a shortage of biotalent at every level, hindering the sector's innovation and development. Given the cross-cutting nature of biotechnology, there is no silver bullet for the talent shortage. Policymakers and industry must convince more students and workers to pursue STEM careers generally, attract those already in STEM tracks to consider biotechnology careers specifically, and broaden pathways into the bioeconomy for those not in STEM careers.

In June 2023, the Biden administration released a plan to expand and diversify America's biotechnology and biomanufacturing workforce, which outlined several initiatives such as prioritizing biomanufacturing in federal support for apprenticeships at the Department of Labor and expanding partnerships with historically Black colleges and universities, tribal colleges and universities,

and minority serving institutions.⁴²⁷ Realizing the full potential of American biopower, however, requires more ambitious steps to build, attract, and retain biotalent.

Supercharge America's biotalent with high-skilled immigration reform

Immigration is one of America's enduring competitive advantages. Global STEM talent wants to study, work, and live in the United States. Political gridlock over comprehensive immigration reform should not impede progress on more targeted, high-skilled immigration reform, given the stakes for U.S. national and economic security. Specifically, Congress should exempt immigrants with STEM PhDs from annual green card caps, dramatically raise the cap on H1-B visas for STEM workers, fund federal immigration agents to clear the visa backlog, and consider eliminating the per-country visa quotas entirely.⁴²⁸ Narrower proposals could include reciprocal, expedited student and work visas for high-tech researchers and entrepreneurs in key national security-relevant fields, including biotechnology, across members of NATO, AUKUS, the Quad, and the Five Eyes alliance.

Industry should expand biotechnology exposure and opportunities in secondary schools

Biotechnology companies and industry associations should proactively engage middle schools and high schools to expose students to exciting biotechnology applications to build interest at an early stage. Firms should also partner with local school districts to create extracurricular activities, internships, and even apprenticeship programs. Beyond this, they could offer guidance on key skills and competencies for students to secure biotechnology careers.

Support greater U.S. participation in biotechnology competitions like iGEM

Industry should partner with secondary and postsecondary schools to create and support teams to compete in the annual iGEM competition to build excitement for synthetic biology among students as they consider careers, especially in historically underrepresented communities and geographies. Local, state, and federal governments should explore mechanisms to provide financial support.

Expand federal initiatives that support biotechnology workforce development

Congress should increase funding for federal initiatives that build domestic biotalent. This includes ensuring robust and sustained funding for NSF, whose grant programs ensure regular funding for biotechnology researchers. Beyond this, NSF has several specific initiatives worth supporting, including:

- The Advanced Technological Education program, which encourages partnerships between secondary and postsecondary education institutions, industry, and state economic development agencies to better educate science and engineering technicians through curriculum development, expanded career pathways, and professional development for secondary and postsecondary faculty.⁴²⁹
- The Experiential Learning for Emerging and Novel Technologies initiative, which promotes partnerships between companies in key emerging technologies, including biotechnology, and workforce development organizations to expand opportunities for people to gain hands-on skills and experiences.⁴³⁰
- The Graduate Research Fellowship Program, NSF's flagship grant that provides a three-year annual stipend of \$37,000, along with \$16,000 for tuition and fees, to outstanding students pursuing STEM masters or doctoral graduate degrees at accredited U.S. institutions.⁴³¹ Current funding only allows for about 2,300 fellowships per fiscal year and is routinely oversubscribed; Congress should double NSF funding to allow roughly 4,500 fellowships annually while specifying that at least 1,000 should focus on biotechnology-related fields.⁴³²

The Department of Commerce, through NIST, funds NIIMBL. Among other goals, this program expands pathways for students from nontraditional backgrounds in careers in biopharmaceutical manufacturing through an intensive, weeklong immersion program.⁴³³ Since its launch in 2019, NIIMBL has served 80 students, with many securing jobs at top biopharmaceutical firms.⁴³⁴ In 2024, the initiative announced its largest cohort yet of 63 students, suggesting considerable opportunity for expansion.⁴³⁵

For its part, NIH has several initiatives to expand the biotechnology workforce, with a focus on biomedical research. Its Undergraduate Scholarship Program provides up to \$20,000 per year of support to

undergraduates interested in fields related to biomedicine, and the Student Intramural Research and Training Award provides stipends to high school, college, and graduate students interested in biomedical research careers.⁴³⁶

Broaden biotalent across the federal government

Like AI, biotechnology's cross-cutting applications require bioliteracy across the federal government. From program managers to procurement officers, intelligence officers, and foreign service officers, effective biotechnology policy will demand far greater biotalent in government. The White House should take several steps, many modeled on the October 2023 AI executive order, to incentivize, expand, and accelerate biotechnology hiring across government, as follows:⁴³⁷

- Create a Biotechnology Talent Task Force under the newly launched National Bioeconomy Board to identify key mission areas and priority competencies for federal biotalent; develop a plan for the rapid recruitment of biotalent in government; track and expedite the hiring of biotechnology talent across the federal government; and identify and disseminate best practices to hire, retrain, and retain biotechnology talent.
- Coordinate agency technology talent programs and fellowship programs, including those described above at NSF and NIH, to maximize federal incentives to expand the biotalent pipeline, with a focus on incentivizing careers beyond traditional biomedical fields.
- Streamline personnel vetting requirements and expedite security clearance reviews, as appropriate, for mission-critical biotechnology positions.
- Consider government-wide direct-hire authority and excepted service appointments, including Schedule A(r) hiring authorities.
- Require relevant agencies to develop internal programs to start familiarizing staff with biotechnology, with opportunities for more advanced training as required by the position.
- Leverage initiatives such as the Presidential Innovation Fellowship and U.S. Digital Service to emphasize candidates with backgrounds in biotechnology.



Unlock Biodata

Biotech leadership hinges on quality biodata. However, the proliferation of biodata in the United States has not been matched by sufficient public and private efforts to clean, coordinate, and analyze them. Tapping into America's biological "big data"—while protecting privacy, security, and public trust—is essential to U.S. biotechnology leadership.

Fully fund the All of Us Research Program, which faced a 34 percent funding cut in FY2024 compared to the previous year

Congress should match NIH's \$541 million request for FY2025 for this project. Strong, long-term funding will ensure not only project completion but also the necessary computing and cybersecurity infrastructure to house, protect, and analyze large-scale data.

Create a secure American Biobank

Biodata has increasingly become a strategic national asset, but it remains fragmented across thousands of public and private entities and is difficult for researchers and developers alike to access. As such, Congress should authorize creation of an American Biobank, modeled on the UK Biobank and structured as an independent nonprofit.⁴³⁸ The American Biobank would be fed initially by government-supported initiatives such as the National Agricultural Library, NIH's All of Us Research Program, and the NCBI, but it would also have the authority to acquire or purchase biodata from other public and private sources that include not only health, but also energy, environmental, and agricultural data. The American Biobank would strive to become the preeminent national repository of biodata. It would serve as a one-stop shop for accredited researchers and small businesses to access both large-scale biological and genetic data in a secure and trusted environment, along with advanced computing power for machine learning analysis. By consolidating significant biodata under a single institution with common, high standards for governance, access, and use rules, a national biobank would help turbocharge U.S. biotechnology research and

innovation while ensuring high standards for privacy and cybersecurity. The UK Biobank began with initial core funding of about \$234 million, derived from both public and private sources.⁴³⁹ Congress should provide roughly half this amount to attract private capital to secure the rest. The envisioned concentration of valuable biological data will of course require an accompanying emphasis on cybersecurity as well as secure access for vetted researchers and ventures.

Alternatively, federal agencies such as HHS can support state governments to create regional biobanks modeled on the success of the Indiana Health Information Exchange, which is currently the “the nation’s largest inter-organizational clinical data repository.”⁴⁴⁰

Clarify and modernize regulations for the use of health and genomic data for biotechnology research

Ambiguity over several existing laws and regulations, namely HIPAA, needlessly hinders biotechnology research. It also encourages data hoarding instead of responsible data sharing. HHS should engage with genomic scientists and issue clear and consistent guidance for how HIPAA’s minimum necessary standard applies to the use of genomic data in certain research activities.⁴⁴¹ This would remove uncertainty, spurring innovation while ensuring compliance with the flagship privacy rule.⁴⁴² At the same time, Congress should modernize laws governing personal medical data to unlock the power of deidentified patient data, which offer benefits for biomedical and genomic research with lower risks to privacy and civil rights.⁴⁴³ Reforms should clarify ownership of deidentified patient data and establish incentives, such as a compensation scheme, for the responsible sharing of deidentified biodata.⁴⁴⁴ However, continued advances in machine learning tools will make it easier to reassociate certain data with individuals, requiring strong, enforceable, and regularly updated technical standards to ensure robust deidentification.⁴⁴⁵

Invest in privacy-preserving and privacy-enhancing technologies and incentivize adoption by biotechnology researchers and health providers

Privacy-enhancing and privacy-preserving technologies, such as differential privacy, homomorphic encryption, and synthetic data, remain in their early stages but offer a promising route for democratic societies to avail

themselves of large-scale biodata without surrendering a commitment to privacy. These techniques remain in varying stages of development, and some—such as homomorphic encryption and synthetic data—will require significant time and investment before they are cost-effective at scale.⁴⁴⁶ In March 2023, the National Science and Technology Council released a *National Strategy to Advance Privacy-Preserving Data Sharing and Analytics* (PPDSA), which outlines several worthy recommendations to accelerate the development and adoption of such technologies.⁴⁴⁷ These include, among many steps, accelerating R&D for current and emerging PPDSA technologies, creating pilot projects for implementing these technologies within the federal government, and establishing technical standards, standardized taxonomies, and testbeds. The administration should actively implement the strategy, and Congress should ensure adequate oversight and resources.

Pursue an all-out effort to make federal biodata AI ready

The federal government has amassed significant biodata housed principally within HHS, NIH, the EPA, and the USDA. However, existing data is often incomplete, inconsistent, outdated, and unstandardized—even within agencies. Significant work remains to clean, curate, and structure the data to prepare it for analysis by advanced machine learning tools. The White House should require select federal agencies to audit their biotechnology-relevant data and require an intensive effort to make it AI ready. At the same time, the White House should develop improved standards to ensure future biotechnology-relevant data collection is compatible with machine learning tools for analysis.

Strengthen cybersecurity at hospitals, clinics, and other biodata repositories

As biological and genetic data assume greater strategic importance, so will ensuring that data remains secure from growing cyber threats. Recent years have brought a rise in ransomware and other cyberattacks against major hospital systems, which often lag in adopting cybersecurity best practices and the latest defensive tools.⁴⁴⁸ According to HHS, there has been a 93 percent increase in cyber breaches reported to its Office for Civil Rights between 2018 and 2022, with a 278 percent increase in large breaches that involve ransomware.⁴⁴⁹ In July 2024, a major ransomware attack targeted OneBlood, a blood donation nonprofit, delaying procedures across

some 300 of its hospital partners.⁴⁵⁰ Last December, HHS released a cybersecurity strategy for the health sector.⁴⁵¹ To help HHS implement the strategy, Congress should provide new funding and authorities for HHS and expand financial and technical assistance programs for U.S. hospitals and clinics, including through its Critical Infrastructure Protection (CIP) program. At a minimum, Congress can provide the \$12 million increase to the CIP that HHS requested for FY2025.

Develop a standardized framework for secure and trusted biodata environments for research

To access secure environments to analyze large-scale, diverse biodata, U.S.-based biotechnology researchers have only a fragmented landscape across government, academic, and private platforms—each with their own access rules, structure, governance, and security practices. The lack of a high, common standard across secure data environments can undermine collaboration, raise costs for researchers to access multiple datasets, and erode public trust. Instead, policymakers should look to the UK’s standardized network of secure data environments, created with an initial investment of nearly \$260 million, which allow vetted researchers to analyze sensitive health and genetic data in a secure, privacy-respecting manner.⁴⁵² HHS should conduct a feasibility study for modeling a similar approach in the United States, which has the potential to create a more unified, consistent, and trustworthy network of large-scale biodatabases.

Accelerate the transition toward a more unified, interoperable system for EHRs and electronic medical records

The archipelago of health providers and lack of consistent, enforceable standards for health and medical data collection has left the United States with a widely fragmented biodata landscape, impeding the nation’s ability to unlock insights from large-scale biodata with advanced machine learning tools. Although the country has made progress toward a more interoperable, standardized ecosystem for biodata—thanks largely to reforms in the 21st Century Cures Act—significant gaps remain. Without reforms to promote EHR standardization, sharing, and interoperability, newly collected biodata will remain largely inaccessible to researchers and developers. One promising initiative is the Trusted Exchange Framework and Common Agreement,

which fosters biodata exchanges through a nationwide network.⁴⁵³ HHS should strengthen the effort by moving from a voluntary to a mandatory participation model for all health providers, expanding its current technical focus on data exchange to include data standardization.



Strengthen and Streamline Biotechnology Policy

As with AI, America must pursue policies that enable cutting-edge bio-innovation, while maintaining safety and security. Biotechnology regulations, to the extent they exist, are outdated, incoherent, and insufficient to provide the stable regulatory framework required for private sector innovation.

With that said, the United States is relatively ahead of some competitors in articulating national priorities and plans for biotechnology and the broader bioeconomy, exemplified by the 2022 Biotechnology EO, the *Bold Goals*, and a flurry of agency-level initiatives. With that said, the United States still lacks significant biotechnology-related legislation to codify a stable and sufficiently resourced national strategy to unlock the full potential of biotechnology. Executive orders do not always survive changing administrations, and even when they do, they are not always prioritized for implementation by successors.

Develop a unified national strategy for American biopower

Despite the flurry of efforts across the administration, the U.S. government still lacks a detailed, long-term strategy to secure America’s place as the preeminent biopower. The 2022 Biotechnology EO remains the closest document to such a strategy. However, it does not set out long-term goals for U.S. biotechnology leadership or identify gaps and synergies across sectors or the civilian and military domains, nor does it include detailed, multiyear plans for implementation and estimated funding needs. Competitors abroad have had detailed biotechnology strategies for years; the next administration should direct the interagency to finally develop a strategy for America.

Strengthen the biotechnology policy architecture across the government

The March 2024 launch of the National Bioeconomy Board was an important and overdue step to elevating biotechnology and improving coordination across the federal government.⁴⁵⁴ The next administration should continue this effort and consider elevating it with the deputy White House chief of staff as chair. Once the administration develops a unified national strategy for American biopower, it should task relevant federal agencies with developing detailed implementation strategies to support its goals through their various policies, programs, investments, and outreach, while identifying agency gaps in resources and authorities for the White House and Congress to review. Relevant agencies should also consider designating a permanent chief biotechnology officer, analogous to the chief AI officer required by the October 2023 executive order on artificial intelligence, with a direct reporting line to at least the undersecretary level.⁴⁵⁵

Update NAICS and NAPCS codes for biotechnologies and the bioeconomy during the 2027 revision process

The 2022 Executive Order on Advancing Biotechnology tasked the DOC with recommending revisions to the NAICS and NAPCS codes ahead of the next five-year review process in 2027. These two federal codes track economic activity across different sectors and products, allowing for consistent measurement across the economy. However, this existing framework focuses on specific sectors and final products, whereas the bioeconomy cuts across multiple sectors and hinges more on the techniques used to create a product, rather than the product itself. As NAICS prepares its 2027 update, it should adopt recommendations from the Interagency Technical Working Group submitted in its September 2023 report.⁴⁵⁶ The next administration should then direct federal agencies to adopt the codes in its work products to ensure consistent analysis across government.

Develop a uniform federal definition for biotechnology and the bioeconomy

Today, different federal agencies maintain definitions of biotechnology and the bioeconomy relevant to their particular jurisdictions.⁴⁵⁷ Beyond this, recent years have seen multiple definitions put forward for the bioeconomy, from the 2020 National Academies

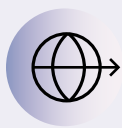
report on *Safeguarding the Bioeconomy* to the 2022 Executive Order on Advancing Biotechnology to the 2022 Bioeconomy Lexicon.⁴⁵⁸ Although each definition captures important nuances, the variance on balance impedes federal coordination, measurement, and policy. It also creates an additional burden for researchers and the private sector—especially smaller biotechnology ventures. Congress should task NIST with developing and proposing uniform federal definitions for biotechnology and the bioeconomy, drawing on the recent lexicon and in coordination with the 2027 NAICS and NAPCS review process.

Establish a single federal office to consolidate biotechnology regulation for nondefense applications

Since 1986, the federal government has sought to streamline biotechnology regulation through the Coordinated Framework. Successive administrations have made incremental progress in increasing the clarity, transparency, and efficiency of the framework that is now divided between the USDA, FDA, and EPA. In May 2024, the three agencies published a detailed plan to further improve the Coordinated Framework in response to the 2022 Executive Order on Advancing Biotechnology.⁴⁵⁹ The plan included many worthy and overdue reforms, such as the USDA's commitment to streamline permit applications and clarify which modified microorganisms are exempt from regulation, and the FDA's pledge to issue clear guidance on its oversight of gene editing products.⁴⁶⁰

Nevertheless, as long as biotechnology regulation remains divided across three large federal agencies, each with their own culture, processes, rules, standards, and bureaucratic interests, it will create needless ambiguity, redundancy, and regulatory burden on biotechnology developers. For years, biotechnology stakeholders have consistently complained that the three agencies often have different definitions, data requirements, and review timelines.⁴⁶¹ As biotechnologies rapidly evolve, it is fair to wonder whether the three leading agencies can proactively update the Coordinated Framework to keep pace.

Alternatively, Congress could replace the Coordinated Framework with a consolidated regulatory office that draws on staff from the three agencies, in addition to outside experts, that can provide a single point of entry for regulation for developers, along with harmonized regulatory definitions, policies, processes, and data requirements, and dedicated staff able to respond promptly and comprehensively.



Lead Globally

Although the United States must initially focus on shoring up its domestic biotechnology ecosystem, its long-term leadership will grow or falter in part based on global engagement to shape norms, secure overseas markets, and spearhead the development of a still-forming international biotechnology architecture. Fortunately, the United States enters the biorevolution with an extraordinary advantage: an unrivaled global network of allies and partners with considerable strengths in biotechnology and biomanufacturing.

European allies such as France, the United Kingdom, Japan, and South Korea are all competitive across a range of biotechnologies, including synthetic biology and genetic engineering. India ranks third after China and the United States in both biological manufacturing and novel antibiotics and antivirals.⁴⁶² More ambitious partnerships could promote greater cooperation around biotalent, joint R&D projects, information sharing on emerging biotech capabilities and supply chains, and efforts to shape norms and standards in line with democratic values.⁴⁶³

The United States should fully exercise its global leadership, and it has little time to waste. As with AI, Washington's decisions in the next few years could determine whether the future of biotechnology will be defined by democratic nations and values, or by an emerging techno-authoritarianism. The stakes are a future of responsibly developed and deployed biotechnologies that unlock positive breakthroughs for human health, energy, agriculture, and defense, or a dystopian world of nonconsensual biosurveillance, reckless bioweapon development, and a frightening race to gain geostrategic advantage through population-wide genetic modifications and human-computer integration.

Convene a global Biotechnology Safety and Opportunity Summit. Modeled after the success of the UK's 2023 Bletchley AI Safety Summit, the United States should convene a Biotechnology Safety and Opportunity Summit to reinforce its global leadership and lead the agenda for the responsible development and deployment of biotechnology.⁴⁶⁴ The summit should strive for a set of high-level commitments, similar to the 2023 Bletchley Declaration, outlining key principles and values for responsible biotechnology research, development, and use.⁴⁶⁵ More ambitiously, it could outline opportunities

for biotechnology-leading nations to partner on joint biotechnology safety research and responsible diffusion to emerging markets. Similarly, the United States should lead an effort through the G7 to articulate principles for responsible biotechnology research, development, and use, modeled after the success of the Hiroshima Process International Guiding Principles for Organizations Developing Advanced AI Systems.⁴⁶⁶

Set a global example by declaring biotechnology red lines

The United States should lead the world by unilaterally clarifying red lines that it will not cross for certain biotechnology applications—for instance, developing bioweapons that target specific populations or essential crops, deploying intrusive biosurveillance platforms, or “designing” human embryos for certain desirable attributes. As the leading biopower, the United States would set a powerful example for other nations to follow by voluntarily renouncing future applications. To develop these standards, the White House should convene a Task Force on Biotechnology Ethics to outline these red lines and articulate principles and suggested federal policies to govern responsible biotechnology research, development, and use across the federal government.

Leverage security partnerships to drive joint biotechnology R&D, investment, and talent exchanges

There is considerable opportunity to elevate biotechnology in long-standing and emerging security partnerships, including NATO, AUKUS, the Quad, and the U.S.–Japan–Republic of Korea trilateral relationship.

To start, the United States should leverage its board representation and financial contribution to NATO's DIANA to prioritize biotechnology, which is now one of several of focus areas. The United States should also modestly contribute to the NIE, which could encourage other alliance members to increase their engagement and help foster a strong transatlantic biotechnology ecosystem.

For AUKUS, the United States should push to add biotechnology to the agenda for Pillar II and integrate it into the newly established AUKUS Advanced Capabilities Industry Forum, AUKUS Defense Investor Network, and trilateral innovation challenges.⁴⁶⁷

In addition, U.S. government officials and companies should continue to mature the QUIN and prioritize biotechnology investments beyond areas of health and medicine, which already receive the overwhelming share of private capital.

Finally, the still-maturing trilateral partnership between the United States, Japan, and the Republic of Korea should launch a trilateral biotechnology industry working group and negotiate a trilateral biotechnology cooperation agreement outlining opportunities for joint investment, trade, and talent exchange, as well as policy and regulatory harmonization.⁴⁶⁸

Develop a strategy for shaping international standards for biotechnology to align with U.S. interests and democratic values

The White House should task the Department of State (DOS), in consultation with cabinets represented on the National Bioeconomy Board, to develop a strategy to shape international standards for biotechnology through the Codex Alimentarius, International Organization for Standardization (ISO), WHO, OECD, and other relevant international bodies, to promote alignment with democratic values, ethics, responsible use, and fair market principles.

Support U.S. biotechnology firms to secure global market share

BGI, WuXi Biologics, and other Chinese biotechnology firms are working to dominate emerging markets to offset anticipated barriers to scaling in developed economies.⁴⁶⁹ Left unchecked, the continued proliferation of Chinese biotechnology firms and products in emerging markets could significantly boost China's biotechnology leadership by increasing overseas revenue available for reinvestment in R&D, expanding and diversifying their available health and genomic data for large-scale analysis, and creating technological path dependency that could confer commercial and geostrategic benefits for years to come. The White House should task the DOS, the U.S. Trade Representative (USTR), and the U.S. Trade and Development Agency with developing a working group with industry to identify priority overseas markets and coordinate diplomatic, financial, and other support to broaden U.S. market share vis-à-vis Chinese firms.

Conclude bilateral and plurilateral digital trade agreements with biotechnology-friendly provisions

The USTR should prioritize negotiation of digital trade agreements with biotechnology-leading allies and partners, similar to the recently concluded Australia-Singapore Digital Economy Agreement, that facilitate

secure exchange of biodata (deidentified as necessary), reduce tariffs and regulatory barriers for biotechnology products, and foster greater harmonization of relevant standards.⁴⁷⁰



Build Public Trust

Strong and sustained U.S. leadership in biotechnology requires enduring support from policymakers and the public. However, both groups remain broadly unaware of the looming biorevolution and its consequences for America's prosperity and security. Considerable work remains to raise public awareness about the biorevolution and build public trust. The botched public communications for OWS—an unambiguous case of American scientific and innovative prowess—missed an opportunity to build public trust in the COVID-19 vaccines and remains a cautionary tale for the cost of neglecting proactive public engagement.⁴⁷¹

Like with AI, biotechnology applications require proactive discussion about risks, ethics, mitigations, and implications for democratic values. The United States should set an example for the world with a proactive debate about responsible research, development, and application of biotechnologies that aligns with a long-standing ethical, legal, and societal implications framework and respects core democratic values while mitigating risks. Publics around the world have reasonable questions, for instance, about the ethics of editing—let alone designing—human and animal genomes or developing treatments to reverse aging or augment human brains with computers. Governments and businesses that sidestep these questions or assume popular support may find their biotechnology ambitions stalled by public opposition. Conversely, businesses and governments now have an opportunity to anticipate these challenges with concerted actions and mitigations that build public awareness, buy-in, and trust.

Hold a national summit on the bioeconomy

Industry should partner with the federal government, along with biotechnology-leading state governments such as Massachusetts, Minnesota, and North Carolina, to

convene a national summit to drive high-level attention to the positive applications of biotechnology, make the case for why biotechnology leadership is essential to U.S. national and economic security, and expand the national debate about responsible research, development, and use.

Senior executive branch leaders should make the case for U.S. biotechnology leadership

High-level executive branch officials, including at a minimum heads of the DoD, DOE, DOC, USDA, Treasury, NSF, HHS, and NIH, along with the national security advisor, should give keynote remarks across the country articulating the opportunities for U.S. biotechnology leadership, from improving defense capabilities and human health to fighting climate change, achieving food and energy security, and creating high-paying jobs across the country—as well as the dangers of ceding that leadership to authoritarian competitors such as China.

Congress should convene regular biotechnology hearings to increase awareness among the public and elected officials about the coming biorevolution

Congress took a major step with the establishment of the NSCEB, whose final report in early 2025 will help drive public and policymaker attention to biotechnology.⁴⁷² Still, relevant congressional committees should mirror the energy they have devoted to AI with regular hearings to explore the economic, security, societal, and other implications of biotechnology for the United States.

Activity on this front has been scant. The Senate Agriculture Committee has not held a biotechnology-focused session since 2015.⁴⁷³ The Senate Committee on Commerce, Science, and Transportation held a subcommittee hearing on biotechnology in 2020, but the full committee has yet to devote one session to this critical area.⁴⁷⁴ Although the Senate Committee on Homeland Security and Governmental Affairs recently advanced the BIOSECURE Act, it also has yet to hold a hearing devoted to biotechnology.⁴⁷⁵

In the House, the Select Committee on the Chinese Communist Party has brought much-needed attention to biotechnology, including through a March 2024 hearing, but other standing committees have yet to devote similar attention.⁴⁷⁶ Congress should drive the national conversation on the opportunities and risks of biotechnology to ensure the public and policymakers are adequately tracking rapid developments in the sector and providing oversight as necessary.

Chapter 06

Conclusion

The biorevolution is here. The question is which nation will lead it, shape it, and harness its power. The country that embraces the biorevolution with ambitious policies and investments will stand the best chance to become the 21st century biopower and secure vast potential benefits for national security and competitiveness.

Stepping back, the United States enters the biorevolution with formidable tailwinds, including its dynamic capital markets, innovative private sector, world-leading higher education institutions, and unrivaled STEM talent. But inaction and complacency are the surest route to surrendering the U.S. edge. To secure America's place as the global biopower, policymakers must take ambitious steps to accelerate tailwinds through an increased investment in broader biotechnology research and applications, an expanded pipeline of biotalent, and a strategy for prioritizing biotechnology in international standards-setting and multilateral partnerships. At the same time, policymakers must navigate headwinds that imperil further progress—specifically, an anti-innovation morass of conflicting standards, definitions, and regulations for biotechnology; inaccessible and insecure U.S. biodata, the fuel for biotechnology innovation; an underdeveloped national infrastructure for biomanufacturing to promote broad-based innovation; and a lack of public awareness and trust in emerging, ethically fraught biotechnology applications such as human gene editing and human-computer interfaces.

If any nation overtakes the United States as the global biopower, it will be China. Significant public investment, long-term policy and prioritization, a massive domestic market, decades of largely unrestricted capital flows and technology transfers, a growing talent base, and amassing of biodata through both licit and illicit means

have helped China close the gap. Beijing already leads high-impact research in biofuels, synthetic biology, and biomanufacturing. Its scholars rank second in the world for authoring biomedical papers. China is a biomanufacturing powerhouse, exporting roughly 40 percent of the world's active pharmaceutical ingredients.⁴⁷⁷ China now aspires to move up the biotechnology value chain and become a biotechnology superpower.

Now is the time for the United States to define a future of biotechnology that unlocks its vast potential for good, while ensuring it develops responsibly, transparently, and in alignment with democratic values.

China's ambition should inspire action from U.S. policymakers to extend and secure America's lead with bold steps to drive research and innovation across the spectrum of biotechnologies, expand American biotalent, responsibly unleash U.S. biodata, leverage global partnerships, and build public trust. Now is the time for the United States to define a future of biotechnology that unlocks its vast potential for good, while ensuring it develops responsibly, transparently, and in alignment with democratic values. The United States led the 20th century in biotechnology, but considerable work remains to secure its position as the global biopower of the 21st.

Appendix: Cost Justification for Recommendations

This table details the costs and justifications for the various recommendations outlined in this report, which total roughly \$20 billion in new federal funding. The amounts only indicate new funding beyond recently enacted levels. Some recommendations are budgeted for a single fiscal year; others are budgeted for up to a decade. Policymakers should consider this the floor of an all-out strategy to secure U.S. biotechnology leadership.

Recommendation	New Funding	Explanation
Preserve and extend the Qualified Advanced Energy Project Credit	\$4 billion	The Inflation Reduction Act allocated \$4 billion for the Qualified Advanced Energy Project Credit. Projects that begin construction by the end of 2024 would be eligible. This would make an additional \$4 billion available to extend the credit for projects that launch by the end of 2026.
Double funding for the DoD's Tri-Service Biotechnology for a Resilient Supply Chain program and Distributed Bioindustrial Manufacturing Program	\$6.35 billion	The Department of Defense (DoD) allocated \$270 million for the Tri-Service Biotechnology for a Resilient Supply Chain program and \$1 billion for the Distributed Bioindustrial Manufacturing Program, for a total of \$1.27 billion over five years starting in 2022. The authors recommend doubling this funding level to \$2.54 billion and extending it at this higher level for an additional five years, for a total of \$6.35 billion in new funding.
Double funding for DARPA's Biological Technologies Office for a decade	\$1.73 billion	The Defense Advanced Research Projects Agency (DARPA) allocated \$345 million in FY2024 for the Materials and Biological Technology Program Element. Congress should double the funding to \$690 million annually—an increase of \$345 million—and sustain this higher funding level for five years.
Fund NIH's FY2025 budget request and specify biotechnology priorities	\$2.4 billion	The National Institutes of Health (NIH) requested \$50.2 billion for FY2025—\$2.4 billion more than its FY2023 appropriation.
Scale the Advanced Research Projects Agency for Health's funding to at least \$3 billion	\$1.5 billion	FY2023 funding was \$1.5 billion. The authors recommend doubling it to \$3 billion.
Build a national biofoundry network	\$750 million	The National Science Foundation (NSF) has made \$75 million a year available for awards. Doubling this amount would raise it to \$150 million annually over 10 years.
Restore funding for NIH's Innovation Account	\$678 million	FY2024 funding was \$407 million, and FY2023 funding was \$1.085 billion. Restoring funding to FY2023 levels would cost \$678 million.
Fund NSF's FY2025 budget request	\$643 million	NSF requested \$10.185 billion for FY2025—\$643 million more than its FY2023 appropriation.
Expand America's Seed Fund and specify a focus on biotechnology	\$500 million	This would allocate \$50 million annually over the next decade for biotechnology-specific ventures.
Increase NSF funding for the Graduate Research Fellowship Program to roughly 4,500 fellowships	\$318.67 million	This would double the FY2023 level of \$318.67 million.
Fund the National Institute for Standards and Technology	\$311.5 million	The National Institute for Standards and Technology requested an additional \$311.5 million for FY2025.
Fund Biotechnology Moonshots and <i>Bold Goals</i>	\$201.54 million	The Department of Energy (DOE) requested \$36.54 million above FY2023 levels for Biological and Environmental Research and \$95 million above FY2024 for the Office of Manufacturing and Energy Supply Chains; the U.S. Department of Agriculture requested \$50 million for the Agriculture Advanced Research and Development Authority (AgARDA); the U.S. Department of Health and Human Services (HHS) requested \$20 million more in FY2025 for the Biomedical Advanced Research and Development Authority (BARDA).
Develop a secure American Biobank	\$116 million	\$116 million is roughly half the initial cost of developing the UK Biobank (\$234 million). This initial public investment could leverage private capital to cover the rest.
Strengthen cybersecurity at hospitals, etc.	\$12 million	HHS requested \$12 million for FY2025 for its Critical Infrastructure Program.
Fulfill the DOE budget requests	\$5 million	The DOE requested \$5 million for FY2025 for its Office of Critical and Emerging Technologies.
Total	\$19.51 billion	

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